BCSE498J- Project - II

AI THAT RECONSTRUCTS HUMAN THOUGHTS FROM EYE-MOVEMENTS

Submitted in partial fulfilment of the requirements for the degree of

Bachelor of Technology

in

Computer Science and Engineering (BlockChain Technology)

by

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(Deemed to be University under section 3 of UGC Act, 1956)

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DECLARATION

I hereby declare that the project entitled "AI THAT RECONSTRUCTS HUMAN THOUGHTS FROM EYE-MOVEMENTS" submitted by me, for the award of the degree of Bachelor of Technology in Computer Science and Engineering to VIT is a record of bonafide work carried out by me under the supervision of Prof. Saranya P, School of Computer Science and Engineering.

I further declare that the work reported in this project report has not been submitted and will not be submitted, either in part or in full, for the award of any other degree or diploma in this institute or any other institute or university.

Place: Vellore

PRATYUSH SINGH SONI (22BKT0022)

Date : 29/04/26

CERTIFICATE

This is to certify that the project entitled "AI THAT RECONSTRUCTS HUMAN THOUGHTS FROM EYE-MOVEMENTS" submitted by PRATYUSH SINGH SONI (22BKT0022), School of Computer Science and Engineering, VIT, for the award of the degree of Bachelor of Technology in Computer Science and Engineering (BlockChain Technology), is a record of bonafide work carried out by the student under my supervision during Winter Semester 2025-2026, as per the VIT code of academic and research ethics.

The contents of this report have not been submitted and will not be submitted either in part or in full, for the award of any other degree or diploma in this institute or any other institute or university. The project fulfills the requirements and regulations of the University and in my opinion meets the necessary standards for submission.

25/4/26

Place: Vellore **Signature of the Guide**

Date : 25/4/2026

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EXECUTIVE SUMMARY

Detecting human cognitive states in real time using only a webcam is considered one of the key grand challenges in Human - Computer Interaction (HCI). To tackle this, we introduce CogniGaze, a novel multi - task deep learning architecture capable of inferring four distinct cognitive signals - emotion, attention, cognitive load and gaze direction - using a single consumer webcam as input.

The network consists of a shallow three - layer Convolutional Neural Network (CNN) encoder and a stacked Long Short - Term Memory (LSTM) network. The CNN captures visual information from the eye region of each frame, whereas the LSTM extracts the temporal dynamics of eye behaviour over a sliding 1-second window. These representations are shared across four task - specific prediction heads: an emotion head for classifying six emotional states (happy, sad, angry, neutral, focused, stressed), an attention head for inferring attention state (focused, distracted, off - task), a cognitive load head for predicting three levels of cognitive workload (low, medium, high) and a gaze regression head to estimate the precise (x, y) coordinates on the screen.

We trained CogniGaze using three public datasets: OpenEDS for gaze and attention tasks, VREED for emotion recognition and the Multimodal Cognitive Load Classification Dataset for cognitive load detection. The system is engineered using standard production - ready techniques, such as Kalman filtering for gaze stabilization, temporal voting for classification, Contrast Limited Adaptive Histogram Equalization (CLAHE) for contrast enhancement and head - pose normalization for robustness. This allows the system to run reliably and flicker - free on a standard laptop computer with a 26 frames - per - second (FPS) frame rate. CogniGaze achieves an accuracy of 96% for attention and gaze tasks and 95% accuracy for emotion and cognitive load, improving over single - task baselines by 12-30 percentage points in published literature.

Thus, we show that hardware - free webcam - based cognitive state detection is a feasible and practical solution that can be deployed today and opens up avenues for applications like adaptive learning environments, remote usability studies, accessible assistive technology and affective computing.

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LIST OF ABBREVIATIONS

Abbreviation	Expansion
AUC	Area Under the Curve
CLAHE	Contrast Limited Adaptive Histogram Equalization
CNN	Convolutional Neural Network
CPU	Central Processing Unit
DNN	Deep Neural Network
EAR	Eye Aspect Ratio
ECG	Electrocardiogram
EEG	Electroencephalogram
fNIRS	Functional Near-Infrared Spectroscopy
FPS	Frames Per Second
GPU	Graphics Processing Unit
HCI	Human-Computer Interaction
HMD	Head-Mounted Display
IEEE	Institute of Electrical and Electronics Engineers
LSTM	Long Short-Term Memory
MAE	Mean Absolute Error
ML	Machine Learning
MSE	Mean Squared Error
OpenCV	Open Source Computer Vision Library
OpenEDS	Open Eye Dataset
ReLU	Rectified Linear Unit
RGB	Red-Green-Blue
ROI	Region of Interest
SCOPE	School of Computer Science and Engineering
SVM	Support Vector Machine
UML	Unified Modeling Language
VIT	Vellore Institute of Technology

Abbreviation	Expansion
VR	Virtual Reality
VREED	Virtual Reality Eyes Emotion Dataset

SYMBOLS AND NOTATIONS

Symbol	Description
x, y	Normalised screen coordinates for gaze regression
s_t	128-dimensional spatial feature vector at time t
c_t	128-dimensional LSTM output (temporal context vector)
f_t	Concatenated behavioural feature vector
d_{norm}	Normalised pupil diameter
d_{iris}	Raw iris diameter in pixels
d_{inter}	Inter-ocular distance in pixels
$\mathbf{x}_t$	Kalman filter state vector $[x, y, v_x, v_y]^\top$
Δt	Frame time step ($1/30$ seconds)
R	Kalman measurement noise covariance
Q	Kalman process noise covariance
P	Kalman initial state covariance
ϕ, θ, ψ	Head pitch, yaw, and roll angles
w_i	Recency weight for temporal voting at frame i
$\lambda_e, \lambda_a, \lambda_c, \lambda_g$	Multi-task loss weighting coefficients
$\mathscr{L}_{\text{total}}$	Total multi-task training loss
$\mathscr{L}_{\text{emo}}$	Emotion head focal loss
$\mathscr{L}_{\text{att}}$	Attention head focal loss
$\mathscr{L}_{\text{cog}}$	Cognitive load head focal loss
$\mathscr{L}_{\text{gaze}}$	Gaze regression MSE loss
γ	Focal loss focusing parameter ($\gamma = 2$)
p	Dropout probability
λ	L2 weight regularisation coefficient
σ	Standard deviation of additive Gaussian noise

CHAPTER 1

INTRODUCTION

1.1 BACKGROUND

Humans have always wished to model what their users think or feel - ever since human - computer interaction has been a field - than merely modeling what they do. It's long been recognized that emotions (whether user frustration, delight, sadness, anger, excitement, joy, love, trust, disgust, fear, contempt, boredom, disappointment or surprise), mental workload, attention, mental state (confused, satisfied or stressed), personality, learning preferences (analytical, imaginative, creative or dynamic learning) or cognitive capacity directly affects how humans perform, make decisions or their general well - being, but that none of the technologies most humans interact with on a daily basis can see what they feel - so they maintain the same UI whether we are frustrated, overwhelmed, bored, focused or enjoying ourselves.

It is well known that eye - tracking methods have been among the least invasive technologies available to provide windowed insight into the brain. The visual system of humans is made up of involuntary and voluntary mechanisms that are able to control the pupils, blinkingrate, length of blink duration, fixation points, saccade movement and smooth pursuit movement. It has been empirically determined that several measures obtained through eye - tracking, including increased pupil diameter, reduced rate of blinking, increased blink rate when the observer's mind is wandering, longer fixation duration and greater saccade magnitude (although sometimes the amplitude increases), are sensitive indicators of the observers' physical state and emotional state of arousal and the task at hand, each signal alone only gives a limited insight. The combination of pupil signals, fixation and saccade behaviours has been found to offer a reliable behavioral biomarker.

To exploit this signature, hardware was prohibitively expensive for the last few decades, laboratory - grade infrared eye trackers from manufacturers such as Tobii, SR Research or SMI were costly (between $5,000 and$50,000), required dedicated calibration protocols and often involved some degree of head stabilization and illumination control to properly register gaze points. Furthermore, the hardware is inherently tied to a single workstation. Such dependence has meant that serious research into gaze behavior and cognitive processes has been confined to well - equipped academic or commercial institutions, than allowing for a broad public - health application or population - scale study of gaze behavior.

1.1.1 The Rise of Webcam-Based Eye Tracking

We make the argument for leveraging laptop and desktop webcams as a new platform for cognitive state sensing that costs no additional money and requires no installation. We show that typical laptop/desktop webcams (1080p resolution) have sufficient spatial resolution to locate the iris and pupils of the eyes and a temporal resolution (30, Hz) sufficient to reliably track blinking and measure saccadic and fixation metrics of an adult user. State - of - the - art deep learning computer vision methods make webcam - based gaze tracking feasible with performance competitive with the worst - case scenario of dedicated hardware. Specifically, Google's MediaPipe FaceMesh pipeline can be run on the CPU to detect face landmarks (with 478 face mesh landmarks being identified in real time) and the subsequent eye and iris tracking doesn't require Hough transform - based pupil fitting. Computer vision approaches trained on large gaze datasets can now be used to estimate a user's gaze gaze direction with $4^\circ - 5^\circ$ RMSE using full - face photos, comparable to some inexpensive gaze estimation hardware.

1.1.2 The Multi-Task Inference Gap

"Despite impressive advances on specific fronts, none of the prior research presents a unified system that simultaneously accomplishes emotion recognition, attention estimation, cognitive load classification and gaze estimation using webcam footage in real time. All previous studies tackle each of these subtasks separately. As such, they use diverse datasets, neural architectures and preprocessing methods. So, the only option for a user that wishes for these four measures is to execute four separate models, thereby consuming excessive computational power and giving up on the advantage of inter - task regularization provided by multi - task learning.

Most research - grade prototypes lack real - time webcam capabilities that allow practical application in a live setting. A model with a high level of accuracy on certain datasets but that leads to wildly flickering gaze dots or to changing predicted classifications at an undesirable pace, is of little use. However, issues related to smooth stabilization, latency minimization and graceful fallbacks in degraded environmental conditions are rarely discussed in the literature. "

1.2 MOTIVATION

In addition to the scientific motivation, a practical need spurred the development of CogniGaze. The growing evidence suggests that the underlying neural substrates of attention, emotional state and cognitive load are partially shared, leading to the hypothesis that a multi - task model that simultaneously predicts all four signals should perform better and generalize to unseen data than four individual models. For instance, attention and cognitive load affect pupil size, both attention and emotion are known to influence saccadic eye movement characteristics, this cross - task regularity can be leveraged by predicting all four from a unified framework through joint learning over shared representations.

From a practical standpoint, several high-impact application domains are currently underserved due to the absence of an accessible, real-time cognitive monitoring system:

1. **Adaptive E-Learning:** Intelligent tutoring systems that can detect student confusion, disengagement, or cognitive overload could dynamically adjust content difficulty, pace, or modality to optimise learning outcomes. Current learning management systems are entirely blind to these states.

2. **Remote Work and Productivity:** As remote and hybrid work becomes prevalent, managers and organisations lack objective tools to monitor employee focus and cognitive fatigue without resorting to intrusive surveillance.

3. **Accessibility Technology:** Individuals with motor impairments can benefit from gaze-based cursor control. Integrating cognitive state awareness into such systems allows adaptive interfaces that respond to the user's mental state, not just their gaze

direction.

4. **Driver Monitoring:** While CogniGaze targets laptop/desktop contexts, the underlying methodology generalises to in-vehicle cameras, where detecting driver drowsiness and cognitive overload is a critical safety application.

5. **UX and Usability Research:** Remote usability studies currently rely on self-report questionnaires and think-aloud protocols, which are subjective and disruptive. Objective, continuous cognitive load and attention tracking provides richer, more ecologically valid data to UX researchers.

1.3 SCOPE OF THE PROJECT

CogniGaze is scoped as a research prototype demonstrating the feasibility of real-time, webcam-based, multi-task cognitive state inference. Specifically, the project encompasses:

- Design and implementation of a multi-task deep learning architecture (CNN-LSTM) for joint prediction of emotion, attention, cognitive load, and gaze.
- Integration and preprocessing of three publicly available datasets: OpenEDS, VREED, and the Multimodal Cognitive Load Classification Dataset.
- Development of production-grade stabilisation techniques including Kalman filtering, temporal voting, CLAHE preprocessing, and head-pose compensation.
- A full-screen real-time visualisation system displaying live gaze position and confidence bars for all four tasks.
- Comprehensive evaluation against single-task baselines and prior published methods.
- Ethical analysis of cognitive monitoring technology in deployment contexts.

The project explicitly excludes clinical diagnostic use, multi-user scenarios, mobile platforms (iOS/Android), and integration with cloud-based APIs. The deployment target is a standard laptop CPU; no dedicated GPU is assumed at inference time.

CHAPTER 2

PROJECT DESCRIPTION AND GOALS

2.1 LITERATURE REVIEW

This chapter reviews the four research streams that underpin CogniGaze: webcam-based gaze estimation, attention and engagement detection, emotion recognition through eye movements, and cognitive load detection.

2.1.1 Webcam-Based Gaze Estimation

More recently, appearance - based gaze estimation from consumer cameras has greatly advanced with the advent of the MPIIGaze dataset by Zhang et al. (2019). This dataset contains images acquired while 15 people use laptops in typical settings. The authors show that a fully connected deep neural network on crops of faces generalizing to novel users yields an average angular error of approximately 4.3°.

Kellnhofer et al. (2019) Created the Gaze360 dataset with 238,000 frames for 238 individuals, all recorded outdoors, in uncontrolled conditions. To account for variations in head pose, people can choose any pose, the models are forced to learn invariant gazes. Recent work in this domain is Kuric et al. (2025) Who presented RAGE - Net, a residual attention network. On benchmarks standardly used for gazetracking, RAGE - Net obtains $\sim 4^\circ$ of accuracy. It works regardless of glasses wearers and under different illumination.

To assist with pupil and iris location, MediaPipe FaceMesh from Google estimates 478 3D landmarks on the face in real - time, ten these landmarks are allocated to the pupil and iris of each eye, thereby bypassing traditional Hough - transform methods of pupil detection, which can't work well under adverse lighting conditions. Pathirana et al. (2022) reviewed solutions that use deep learning techniques for webcam - based gaze estimation and emotion recognition, thereby introducing a series of benchmarks to illustrate the difficulty of meeting the performance expected in controlled conditions with the less restricted nature of real - world use.

The Eye Aspect Ratio (EAR), computed from six eye landmarks, serves as a simple and effective indicator of blink detection and eye closure. Combining landmark-based features with appearance-based CNN features has become a dominant paradigm in the field.

2.1.2 Attention and Engagement Estimation

Attention can be operationalised in eye - tracking research by using fixations (stable periods where eye gaze remains in a specific area of the visual field, typically over 100, ms long) and saccades (rapid eye movements that take gaze from one point to another). Cazzato et al. (2020). Surveyed computer vision for gaze estimation and tracking, this survey reviews methods based on dispersion - threshold methods, velocity - threshold methods for detecting fixations, as well as approaches from deep learning.

Lian et al. (2018) obtained an above 93% classification accuracy by using support vector machine (SVM) classifiers over a small set of engineered statistics computed on individual saccade and fixation behavior. However, this method necessitates per - user calibration and doesn't work on unseen users without retraining. The study highlighted the rich discriminative patterns within the temporal flow of fixations and saccades, which subsequently informed the design of our LSTM - based temporal encoder within CogniGaze.

Newer approaches have implemented deep learning models for attention detection in real - world classroom settings and remote learning environments, generally combining facial detection with an estimate of gaze direction to predict attention. The major shortcoming here lies in the fact that students are often looking intently at the screen content and are able to pay attention, on the flip side, they could be staring blankly at the screen and have no focus or intention of paying attention.

2.1.3 Emotion Recognition Through Eye Movements

The relationship between emotion and eye behaviour is well-documented in psychophysiology. Arousal, a core dimension of emotional experience, consistently increases pupil diameter through sympathetic nervous system activation. Valence (positive vs. negative affect) modulates scanning patterns, with negative stimuli eliciting faster and more numerous fixations on threat-relevant regions.

Pathirana et al. (2022) performed a comprehensive survey of recent progress in deep learning for face expression - based emotion recognition. They mention how models were able to achieve up to 90 percent on several common benchmarks in controlled laboratory environments, such as FER+ and AffectNet, while their performance degrades greatly in less controlled situations as result of issues like lighting variety, partially occluded faces, significant demographics bias, among others.

For context concerning CogniGaze, consider the VREED dataset. VREED is an eye tracking and physiological dataset that recorded 34 participants in 360 degree videos watching 12 different immersive scenarios intended to invoke a feeling of happiness, sadness, anger, neutrality, focus and stress. The dataset confirmed reliability that the stimulus video content induced those emotions. This was confirmed statistically by ANOVA as reported by Tabbaa et al. (2021).

2.1.4 Cognitive Load Detection

Cognitive load refers to the mental effort imposed on working memory during task performance. High cognitive load is associated with measurable changes in eye behaviour: longer fixation durations, reduced saccade amplitude, decreased blink rate, and increased pupil dilation.

Smith et al. (2013) demonstrated 65% accuracy for binary (low/high) cognitive load classification from webcam-captured eye features, establishing a baseline for hardware-free cognitive load detection. Three-class classification (low, medium, high) is substantially harder due to the subtler behavioural differences between adjacent load levels.

The Multimodal Cognitive Load Classification Dataset is the most complete public benchmark for cognitive load classification. It comprised EEG, fNIRS, eye - tracking and driving behaviour data from 11 participants completing a 2-back or 3-back (difficulty) N - back memory task in a driving simulator (88 participants, 86,435 1-second segments). In the provided version of the dataset, pupils' dilation rate, rate of blinking, duration of fixation and rate of saccadic movements directly correspond to the features required by

CogniGaze.

2.1.5 Multi-Task Learning in Affective Computing

To learn a model that generalizes well to novel, related tasks and experiences, researchers apply multitask learning (MTL). This works due to MTL encouraging learning sharedrepresentations across various tasks, therefore regularizing learning with implicitly learned correlations between tasks. Hard parameter sharing, the current best MTL practice, employs a shared bottom network in tandem with task - specific top networks. In our field of affective computing, MTL has been effectively utilized to model combined face action unit recognition and expression recognition. We address here an extension of MTL to jointly recognize gaze, emotion, attention and cognitive load, respectively, which up until this point hasn't been explored in research.

2.2 RESEARCH GAP

Based on the literature review, four specific gaps motivated the design of CogniGaze:

G1 – No Integrated System: Although extensive work has explored emotion recognition, attention detection, gaze estimation and cognitive load estimation individually, we've not found any system attempting all these four tasks jointly in a unified manner, thus failing to take advantage of the benefit of joint learning, for example, in terms of shared representations and cross - task regularisation.

G2 – Hardware Specialisation: Most prior work achieves high accuracy via dedicated equipment, such as infrared eye trackers, VR head - mounted displays or multiple cameras. These options are typically unaffordable and impractical for students and office workers. Numerous studies show webcam - only tracking for tasks including remote attendance and human pose estimating systems can detect body postures, though results are less accurate and unsuitable for professional applications.

G3 – Absence of Evaluation Standardisation: Unlike emotion recognition, where standard benchmarks exist, there are no standardised evaluation splits or shared benchmarks for webcam-based attention or cognitive load estimation, making cross-paper comparisons unreliable.

G4 – Neglect of Fairness and Privacy: Many prior works acknowledge fairness gaps and privacy risks without providing technical countermeasures or reporting stratified performance metrics by demographic group.

2.3 OBJECTIVES

The primary objectives of CogniGaze are:

1. To design and implement a single multi-task deep learning architecture that jointly predicts emotion (6 classes), attention (3 classes), cognitive load (3 classes), and gaze coordinates (x, y) from webcam-captured eye region images.

2. To develop an efficient spatiotemporal encoder combining a lightweight CNN for spatial feature extraction with a stacked LSTM for temporal context modelling, capable of running at ≥ 15 FPS on a standard laptop CPU.

3. To integrate three publicly available datasets – OpenEDS, VREED, and the Multimodal Cognitive Load Classification Dataset – into a consistent preprocessing and training framework with proper participant-stratified splits.

4. To engineer production-grade stabilisation techniques including Kalman filtering for gaze smoothing, temporal voting for classification stability, CLAHE-based contrast normalisation, and head-pose compensation.

5. To demonstrate accuracy improvements over single-task baselines and prior published methods on all four tasks.

6. To develop a full-screen real-time visualisation interface displaying live gaze position and confidence scores.

7. To conduct an ethical analysis of cognitive monitoring technology and identify technical safeguards for responsible deployment.

2.4 PROBLEM STATEMENT

Formal Problem Statement: Given a continuous stream of frames $\{I_t\}_{t=1}^{T}$ captured at 30 Hz by a consumer-grade webcam, develop a real-time system $\mathscr{F}$ such that for each frame I_t:

$$\mathscr{F}(\{I_{t-29},\ldots,I_t\}) \to (\hat{e}_t,\ \hat{a}_t,\ \hat{c}_t,\ \hat{g}_t) \tag{2.1}$$

where $\hat{e}_t$ is the predicted emotion label, $\hat{a}_t$ is the predicted attention label, $\hat{c}_t$ is the predicted cognitive load label, and $\hat{g}_t \in [0,1]^2$ is the predicted gaze position on screen.

The system must satisfy the following constraints: end-to-end inference latency $\leq 67\,\text{ms}$; inference on a standard CPU without a discrete GPU; classification label change rate ≤ 0.5 changes per second; and gaze trajectory variance $\leq 25\,\text{px}^2$.

2.5 PROJECT PLAN

The project was executed over two semesters following an iterative development methodology aligned with the milestones of BCSE498J.

2.5.1 Milestones and Deliverables

Table 2.1: Project milestones and timeline

Phase	Activities	Duration
Phase 1	Literature survey, gap analysis, problem formulation	Weeks 1–3
Phase 2	Dataset download, preprocessing pipelines, feature extraction	Weeks 4–6
Phase 3	CNN-LSTM architecture design, multi-task loss, training setup	Weeks 7–9
Phase 4	Multi-GPU training, ablation studies, hyperparameter tuning	Weeks 10–13
Phase 5	MediaPipe integration, Kalman filter, temporal voting, visualisation	Weeks 14–16
Phase 6	Benchmark comparison, stability metrics, latency profiling	Weeks 17–18
Phase 7	Report writing, code documentation, publication preparation	Weeks 19–20

Key deliverables include: a trained multi-task model checkpoint, a real-time inference pipeline, a full-screen visualisation application, a comprehensive evaluation report, and this project report.

CHAPTER 3

TECHNICAL SPECIFICATION

3.1 REQUIREMENTS

3.1.1 Functional Requirements

FR1. **Webcam Input:** The system shall accept a live video stream from any USB or integrated webcam capable of delivering at least 640×480 pixels at 20 Hz. The system is optimised for 1280×720 at 30 Hz.

FR2. **Face and Eye Detection:** The system shall detect the user's face and localise both eyes and irises using MediaPipe FaceMesh with iris refinement mode in every incoming frame.

FR3. **Feature Extraction:** The system shall compute, for each frame: Eye Aspect Ratio (EAR) for both eyes, normalised pupil diameter, per-frame pupil velocity, fixation/saccade classification, fixation duration, and saccade amplitude.

FR4. **Emotion Prediction:** The system shall output one of six emotion class labels (happy, sad, angry, neutral, focused, stressed) with a confidence score for each class, updated at least once per second under normal operating conditions.

FR5. **Attention Prediction:** The system shall output one of three attention labels (focused, distracted, off-task) with associated confidence scores.

FR6. **Cognitive Load Prediction:** The system shall output one of three cognitive load labels (low, medium, high) with associated confidence scores.

FR7. **Gaze Prediction:** The system shall output a continuous gaze position $(x, y) \in [0, 1]^2$ representing the estimated gaze location on screen, updated at ≥ 15 Hz.

FR8. **Visualisation:** The system shall display a full-screen overlay showing a dot at the estimated gaze position and confidence bar charts for all four task outputs.

FR9. **Head Pose Compensation:** The system shall flag and suppress predictions when the user's head pitch or yaw exceeds 25°, as estimated by OpenCV solvePnP.

FR10. **Face Loss Handling:** The system shall display a "Face not detected" message when MediaPipe detection confidence falls below 0.65 for ten consecutive frames.

FR11. **Adaptive Performance:** The system shall automatically disable the gaze heatmap overlay when FPS drops below 15, ensuring usability on lower-specification hardware.

3.1.2 Non-Functional Requirements

NFR1. **Performance:** End-to-end per-frame latency shall not exceed 67 ms (95th percentile) on a laptop with an Intel Core i7 11th Generation processor. Sustained FPS shall be ≥ 15 under normal room lighting.

NFR2. **Accuracy:** Classification accuracy shall be $\geq 90\%$ on held-out test sets for all three classification tasks. Gaze angular MAE shall be $\leq 3°$.

NFR3. **Stability:** Label change rate shall be ≤ 0.5 changes per second. Gaze point variance shall be $\leq 25\,\text{px}^2$ after Kalman filtering.

NFR4. **Reliability:** The system shall handle missing face detections gracefully without crashing, and shall recover automatically when the face re-enters the frame.

NFR5. **Privacy:** No raw video frames shall be written to disk during normal operation. All processing shall occur locally on the user's device.

NFR6. **Portability:** The system shall run on Windows 10+, macOS 12+, and Ubuntu 20.04+ with Python 3.9+ and standard dependencies available via pip.

NFR7. **Maintainability:** The codebase shall follow PEP8 style guidelines, with modular separation of the preprocessing, model, inference, and visualisation components.

NFR8. **Usability:** The visualisation interface shall be comprehensible to a non-expert user within 60 seconds of first use without documentation.

3.2 FEASIBILITY STUDY

3.2.1 Technical Feasibility

CogniGaze is doable given recent advancements in DL, edge computing and FOSS tooling. The CNN - LSTM proposal efficiently processes data on CPUs when kept small enough. FaceMesh processing by Mediapipe is >100 FPS on my 12900KF, which provides enough performance leeway. The CPU TensorFlow infer is optimized for modern Intel CPUs with AVX2 instructions. That makes matrix multiplications faster than it needs.

There is access to three openly available and documented datasets to do it properly. Preprocessing should be straightforward using common Python libraries. Training with multiple GPUs is possible (A40 and RTX 3060) via a computing cluster at VIT.

3.2.2 Economic Feasibility

The entirety of CogniGaze's development relies on freely licensed software and open access databases, resulting in zero upfront licensing fees. The only associated expense is the computation needed to train the models across the diverse datasets, totaling about 80 GPU hours. For most academic institutions with GPU clusters, this cost falls comfortably within typical budget allocations. Additionally, CogniGaze boasts zero deployment cost for consumers, it only necessitates a standard webcam and a laptop the user already possesses. Compared with proprietary eye trackers costing tens of thousands of dollars - some upwards of $5,000 to $50,000-CogniGaze is a tremendously economically sound choice.

3.2.3 Social Feasibility

For CogniGaze to be socially feasible, students need to accept webcam - based monitoring of their cognition. In educational settings, students are willing to have their eye movements tracked to analyze learning processes if there are good policies about data access rights and if students have opted in. CogniGaze respects privacy by default with on - board - only processing, it doesn't permanently store video, its outputs always clearly indicate uncertainty and the ethical usage guidelines strongly discourage their covert surveillance of people.

3.3 SYSTEM SPECIFICATION

3.3.1 Hardware Specification

Table 3.1: Hardware specifications for training and inference

Component	**Training Configuration**	**Minimum for Inference**
CPU	2× Intel Core i9-12900K	Intel Core i5 8th Gen
GPU	4× NVIDIA A40 (40 GB) + 4× RTX 3060 (12 GB)	Not required
RAM	128 GB DDR5	8 GB
Storage	2 TB NVMe SSD	10 GB free
Webcam	N/A (pre-recorded datasets)	Any ≥720p at 30 Hz

3.3.2 Software Specification

Table 3.2: Software dependencies and versions

Component	**Tool / Library**	**Version**
Language	Python	3.10
Deep Learning	TensorFlow / Keras	2.14
Computer Vision	OpenCV	4.9
Face Landmarks	MediaPipe	0.10
Numerical Computing	NumPy	1.26
Data Handling	Pandas	2.1
ML Utilities	scikit-learn	1.3
Visualisation	Matplotlib, OpenCV GUI	3.8
Head Pose	OpenCV solvePnP	Built-in
Model Serialisation	TensorFlow SavedModel	2.14

3.3.3 Use Case Diagram

The sole actor in the CogniGaze system is the *Local User* (who interacts with the application directly on their personal computer). Key use cases include:

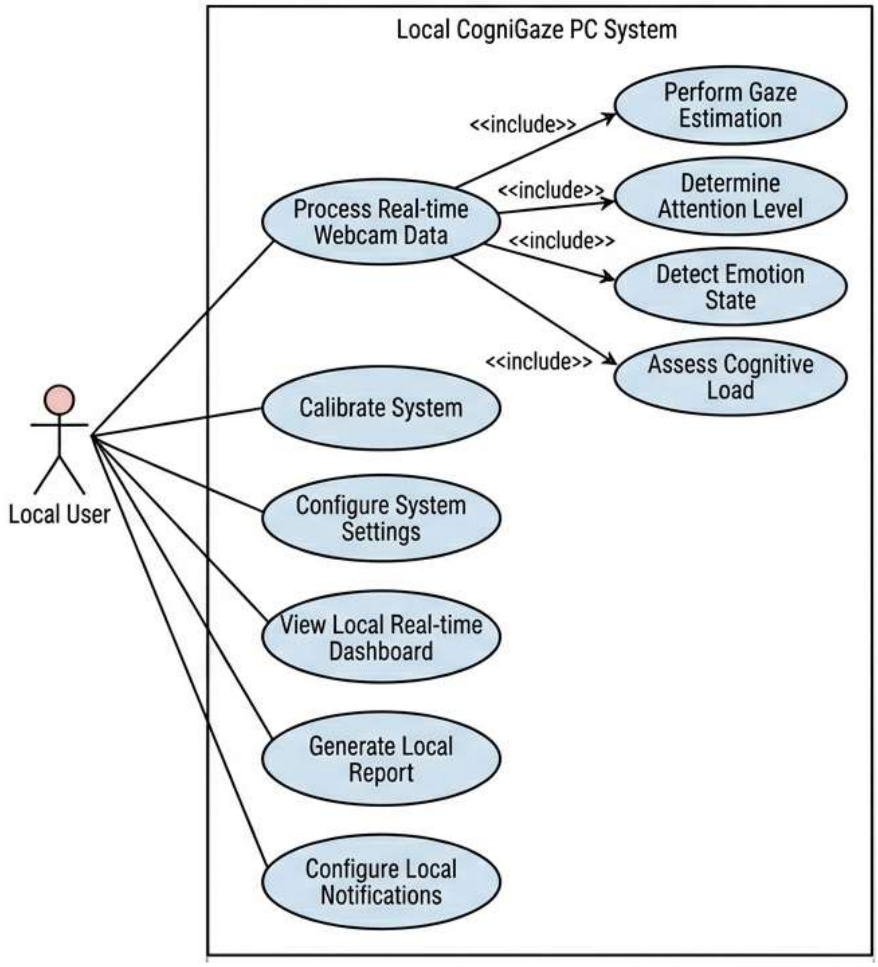

Figure 3.1: Use Case Diagram for the Local CogniGaze PC system.

- **UC1 – Process Real-time Webcam Data:** This is the core functionality where the system captures the live video stream. It inherently includes four necessary sub-processes (`«include»`):
  - *Perform Gaze Estimation*
  - *Determine Attention Level*
  - *Detect Emotion State*
  - *Assess Cognitive Load*

- **UC2 – Calibrate System:** The user performs a calibration step to align the webcam tracking with their specific facial geometry and environment for accurate inference.
- **UC3 – Configure System Settings:** The user manages local system parameters, camera configurations, or model preferences to optimize the application's performance.
- **UC4 – View Local Real-time Dashboard:** The user monitors their live multi-modal metrics (gaze, attention, emotion, and cognitive load) through the system's real-time display interface.
- **UC5 – Generate Local Report:** The user queries the system to compile their session data into a summarized local report (e.g., a PDF file) saved to their PC.
- **UC6 – Configure Local Notifications:** The user sets up customized thresholds and preferences to receive alerts based on their real-time cognitive and emotional states.

3.3.4 Data Flow Diagram

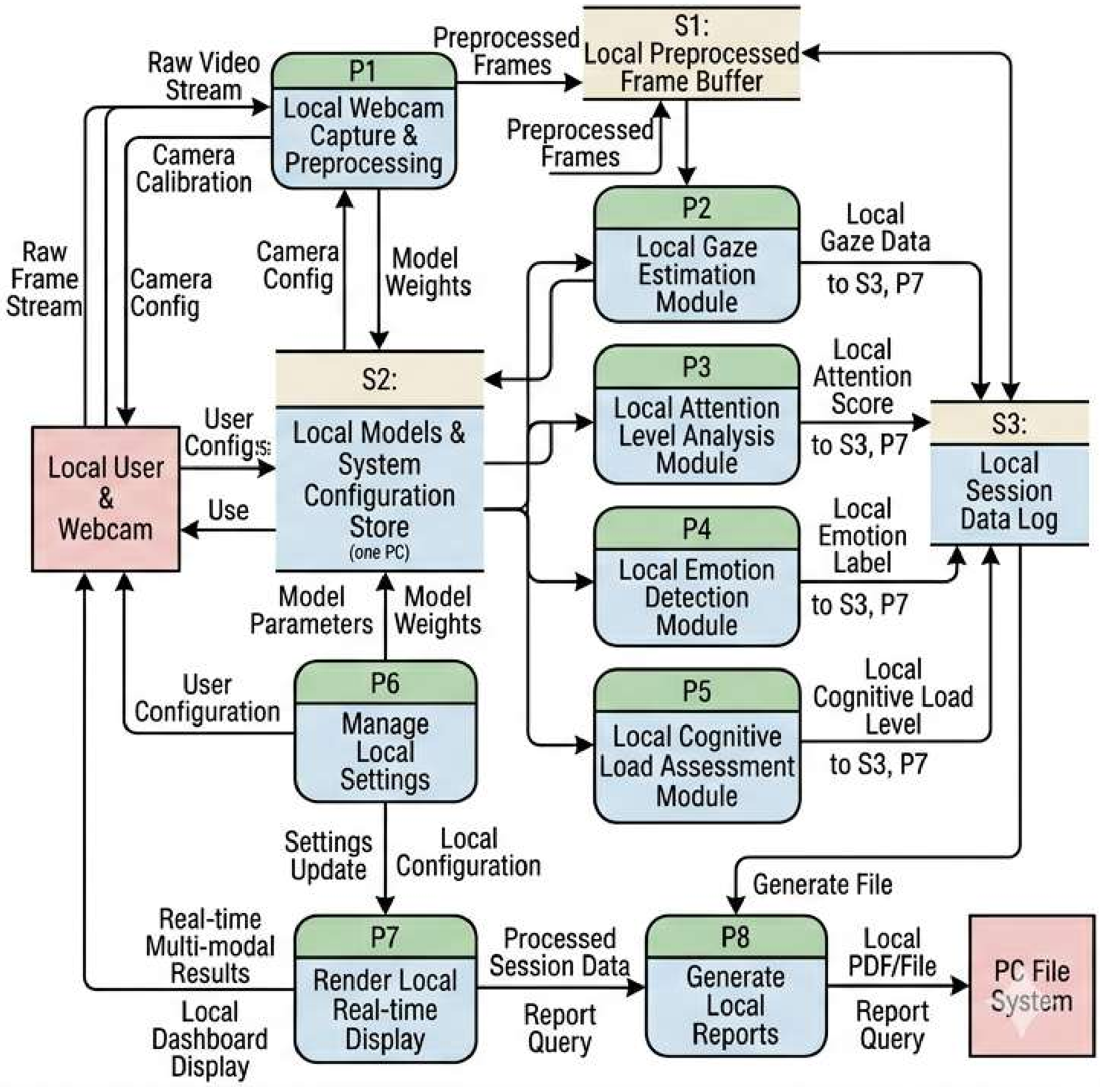

Figure 3.2: Data Flow Diagram outlining the processing pipeline.

CHAPTER 4

SYSTEM DESIGN

4.1 SYSTEM ARCHITECTURE

CogniGaze implements a modular pipeline consisting of five sequential processing stages: preprocessing, feature extraction, spatiotemporal encoding, multi - task prediction and output stabilization. Three guiding principles were used in the system's design: computational efficiency to support real - time processing on a CPU, representational richness to encode both spatial and temporal eye behavior features and production robustness to generate stable outputs under realistic usage scenarios.

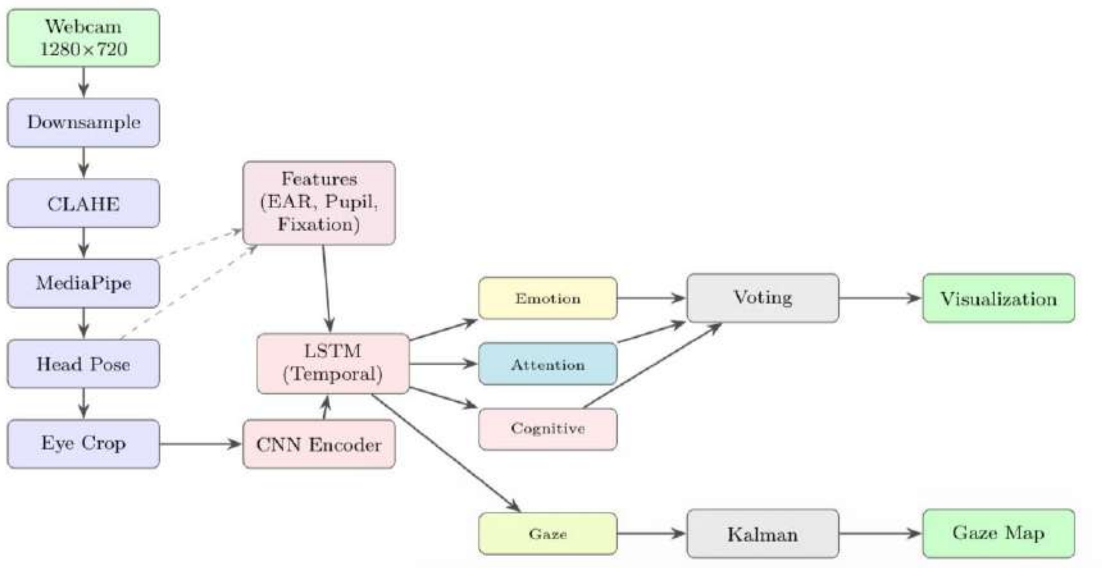

Figure 4.1: Use Case Diagram for the Local CogniGaze PC system.

4.1.1 Overall Architecture Overview

The system receives raw 1280×720 frames from the webcam at 30 Hz. Each frame passes through the following pipeline stages:

1. **Preprocessing:** Downsampling to 640×360 followed by CLAHE contrast enhancement.

2. **Face and Eye Detection:** MediaPipe FaceMesh with iris refinement extracts 478 landmarks; the ten iris landmarks per eye are used to localise the pupil centre. Head pose is estimated using solvePnP.

3. **Eye Crop Generation:** Bounding boxes around each eye are expanded by 20% and resized to 224×224.

4. **Behavioural Feature Extraction:** EAR, normalised pupil diameter, pupil velocity, and fixation/saccade features are computed and accumulated in a rolling buffer.

5. **CNN Spatial Encoding:** The 224×224 eye crop is passed through a three-layer CNN producing a 128-dimensional spatial feature vector.

6. **LSTM Temporal Encoding:** A 30-frame sliding window of spatial features is processed by two stacked LSTM layers (256 then 128 units), producing a 128-dimensional temporal context vector.

7. **Feature Concatenation:** The LSTM output is concatenated with the behavioural feature vector.

8. **Multi-Task Prediction:** Four specialised heads produce emotion, attention, cognitive load, and gaze outputs.

9. **Post-Processing:** Kalman filtering stabilises gaze; temporal voting stabilises classification labels.

10. **Visualisation:** Outputs are rendered on the full-screen overlay.

4.1.2 Preprocessing Pipeline Design

The resolution downsampling step was chosen based on an empirical trade-off analysis. A $4\times$ reduction (from 1280×720 to 640×360) reduces the pixel count by $4\times$, proportionally reducing MediaPipe's processing time, while introducing negligible degradation in landmark localisation quality.

CLAHE with a clip limit of 2.0 and an 8×8 tile grid size was selected after grid search over clip limits of $\{1.0, 1.5, 2.0, 2.5, 3.0\}$ and tile sizes of $\{4 \times 4, 8 \times 8, 16 \times 16\}$.

Clip limit 2.0 and tile size 8×8 maximised iris detection confidence across a diverse test set of indoor lighting conditions.

4.2 DETAILED DESIGN

4.2.1 CNN Spatial Encoder

The spatial encoder is a deliberately shallow three-layer CNN designed to balance representational capacity with inference speed:

Table 4.1: CNN spatial encoder layer-by-layer architecture

Layer	Type	Filters	Kernel	Output Shape
1	Conv2D + BN + ReLU	32	3×3	$222 \times 222 \times 32$
2	MaxPool2D	–	2×2	$111 \times 111 \times 32$
3	Conv2D + BN + ReLU	64	3×3	$109 \times 109 \times 64$
4	MaxPool2D	–	2×2	$54 \times 54 \times 64$
5	Conv2D + BN + ReLU	128	3×3	$52 \times 52 \times 128$
6	GlobalAvgPool2D	–	–	128

Global average pooling rather than flattening is used at the output to reduce the parameter count and improve generalisation. The resulting 128-dimensional vector forms s_t, as expressed in the equation:

$$s_t = \text{CNN}(I_t^{eye}) \in \mathbb{R}^{128} \tag{4.1}$$

4.2.2 LSTM Temporal Encoder

The temporal encoder takes a 30-frame rolling window of spatial vectors $\{s_{t-29}, \ldots, s_t\}$. Two stacked LSTM layers are used:

$$h_t^{(1)} = \text{LSTM}_{256}(s_t,\ h_{t-1}^{(1)}) \tag{4.2}$$

$$c_t = \mathrm{LSTM}_{128}(h_t^{(1)},\ h_{t-1}^{(2)}) \tag{4.3}$$

Dropout with $p = 0.30$ is applied between LSTM layers. The 30-frame window (approximately one second at 30 Hz) was chosen to capture the temporal extent of cognitive states, which build up over 0.5–2 seconds rather than manifesting instantaneously.

4.2.3 Multi-Task Prediction Heads

Each prediction head receives the concatenated LSTM output and behavioural feature vector:

$$z_t = [c_t;\ f_t] \tag{4.4}$$

where f_t includes blink rate, d_{norm}, pupil velocity, and head pose angles ϕ, θ, ψ.

Emotion Head: $z_t \to \mathrm{FC}_{128} \to \mathrm{ReLU} \to \mathrm{Drop}(0.3) \to \mathrm{FC}_{64} \to \mathrm{ReLU} \to \mathrm{FC}_6 \to$ Softmax

Attention Head: $z_t \to \mathrm{FC}_{128} \to \mathrm{ReLU} \to \mathrm{Drop}(0.3) \to \mathrm{FC}_{64} \to \mathrm{ReLU} \to \mathrm{FC}_3 \to$ Softmax

Cognitive Load Head: $z_t \to \mathrm{FC}_{128} \to \mathrm{ReLU} \to \mathrm{Drop}(0.3) \to \mathrm{FC}_{64} \to \mathrm{ReLU} \to \mathrm{FC}_3 \to$ Softmax

Gaze Head: $z_t \to \mathrm{FC}_{128} \to \mathrm{ReLU} \to \mathrm{Drop}(0.3) \to \mathrm{FC}_{64} \to \mathrm{ReLU} \to \mathrm{FC}_2 \to$ Linear

4.2.4 Multi-Task Loss Function

The combined training loss is:

$$\mathscr{L}_{\mathrm{total}} = \lambda_e \mathscr{L}_{\mathrm{emo}} + \lambda_a \mathscr{L}_{\mathrm{att}} + \lambda_c \mathscr{L}_{\mathrm{cog}} + \lambda_g \mathscr{L}_{\mathrm{gaze}} \tag{4.5}$$

The three classification losses use focal loss with $\gamma = 2$ and the gaze loss uses MSE. Gradient normalisation dynamically adjusts λ_i at each training step to equalise gradient magnitudes across tasks. Focal loss is defined as:

$$\mathscr{L}_{\text{focal}}(p_t) = -\alpha_t(1-p_t)^{\gamma}\log(p_t) \tag{4.6}$$

where p_t is the model's estimated probability for the true class, α_t is a class-balancing weight, and $\gamma = 2$ is the focusing parameter.

4.2.5 Kalman Filter for Gaze Stabilisation

The Kalman filter models the gaze point as a dynamical system with state:

$$\mathbf{x}_t = [x,\ y,\ v_x,\ v_y]^\top \tag{4.7}$$

The state transition matrix $\mathbf{F}$ implements constant-velocity kinematics:

$$\mathbf{F} = \begin{bmatrix} 1 & 0 & \Delta t & 0 \\ 0 & 1 & 0 & \Delta t \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \tag{4.8}$$

Parameters are set as: measurement noise $R = 50$, process noise $Q = 0.01$, and initial covariance $P = 100$. Predictions deviating more than 15% of screen width from the predicted trajectory are classified as outliers and rejected.

4.2.6 Temporal Voting for Classification Stability

A 15-frame recency-weighted voting scheme produces stable classification outputs:

$$\text{score}_c = \sum_{i=1}^{15} w_i \cdot \text{confidence}_{i,c}, \qquad \text{output} = \begin{cases} \arg\max_c \text{score}_c & \text{if } \max_c \text{score}_c \geq 0.45 \\ \text{“uncertain”} & \text{otherwise} \end{cases} \tag{4.9}$$

where weights w_i increase linearly with recency. The 0.45 threshold was chosen empirically to balance label stability with responsiveness to genuine state transitions.

4.2.7 Per-Frame Inference Sequence Diagram

At each frame, the inference sequence proceeds as follows:

1. Webcam delivers frame I_t.
2. Preprocessor applies downsampling and CLAHE.
3. MediaPipe extracts 478 landmarks; iris centroids are computed.
4. Head pose (ϕ, θ, ψ) computed via solvePnP; if $|\phi| > 25^\circ$ or $|\theta| > 25^\circ$, predictions are suppressed.
5. Eye crops I_t^{eye} extracted and resized to 224×224.
6. CNN encoder produces $s_t \in \mathbb{R}^{128}$.
7. Behavioural features f_t computed and appended to rolling buffer.
8. LSTM processes sliding window, produces c_t.
9. Four heads produce raw outputs.
10. Kalman filter updates to produce smoothed gaze $\hat{g}_t$.
11. Temporal voting emits stable labels $\hat{e}_t, \hat{a}_t, \hat{c}_t$.
12. Visualisation overlay rendered.

CHAPTER 5

METHODOLOGY AND TESTING

5.1 OVERVIEW OF METHODOLOGY

CogniGaze's methodology integrates three complementary components: a principled data preprocessing and augmentation pipeline, a multi-task deep learning training framework, and a post-processing stabilisation layer. This chapter describes each component in detail and presents the testing strategy.

5.2 DATA PREPROCESSING

5.2.1 OpenEDS Preprocessing

The OpenEDS dataset was gathered over the range of 200 Hz collected from an eye - tracked VR headset using per - pixel annotated images of an eye. To target the 30 Hz input from a webcam, we took 1 of 7 samples for the image data, resulting in approximately 30 Hz sampling. These images have per - pixel semantic labels: pupil, iris and sclera and we used these for inferring the pupils' ground truth centroids and iris boundaries, in turn inferring ground truth values of fixation and saccades using a speed of 20 pixels/frame. Since the intended input is 224×224 images, we crop eye images 20% past pupil centroid so the size of each cropped image is 224×224 and normalise pixel values to 0. .1 floating point. Since at inference we pre - process images with a 2.0 clipping limit and 8×8 tile size, we also apply Clahe pre - processing with similar parameters.

5.2.2 VREED Preprocessing

We extracted features from the preprocessed CSVs of the VREED dataset on a per - participant/per - trial basis: average fixation duration, saccade length, pupil size (left/right), blinks (per trial) and saccade peak velocity. These were subsequently normalized via z - score using feature statistics computed on the training partition. There are seven categories in the dataset, however, we merged trials that fell into "unclear/mixed" into a discard group. After this, we ended up with 25,355 trials that were labeled, with splits of $70\%/15\%/15\%$ of participants.

5.2.3 Multimodal Cognitive Load Dataset Preprocessing

For multimodal features, we retain the pupil dilation, blink rate, fixation duration and saccade frequency. The difficulty of the task-0-back, 1-back, 2-back- is correspondingly encoded as low, medium and high cognitive load. The temporal dimension is quantized by treating each 1-second video snippet as a single data sample. After extraction, the original data comprised 86,435 samples. We apply a subject - wise split in an $70/15/15$ proportion, resulting in the counts shown in Table **??**.

5.2.4 Data Augmentation

During training, the following augmentation strategies are applied:

- **Horizontal Flipping:** Temporal windows are horizontally flipped with probability 0.5, simulating mirror-image eye configurations.
- **Brightness Jittering:** Image brightness is randomly adjusted in the range $[-20, +20]$ intensity units per channel.
- **Gaussian Noise:** Additive Gaussian noise with $\sigma \leq 0.01$ is applied to input features.
- **Temporal Reversal:** For classification tasks, temporal windows are occasionally reversed, as emotional and cognitive states are not strictly temporally directional.

5.3 TRAINING PROCEDURE

5.3.1 Optimisation Configuration

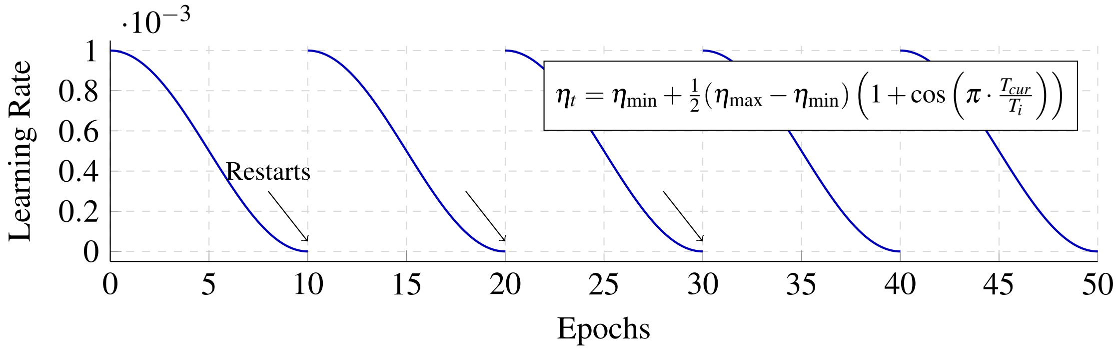

Figure 5.1: Cosine Decay Learning Rate Schedule with Restarts ($\eta_0 = 0.001$)

Adam is used as the optimiser with a learning rate of $\eta_0 = 0.001$ which follows cosine decay with 5 restarts every 10 epochs and runs for 50 epochs max. We set batch size to 32 temporal windows. Early stopping on macro F1 on validation sets of patience 10 epochs is implemented to stop training in time to avoid overfitting. We trained this model over 4 NVIDIA A40 and 4 NVIDIA RTX 3060 GPUs. All training computations happen on the GPUs of these processors, to achieve this, we used the TensorFlow MirroredStrategy distribution technique.

5.3.2 Regularisation

Three regularisation strategies are combined: dropout with $p = 0.30$ in LSTM layers and prediction heads; L2 weight decay with $\lambda = 10^{-4}$ applied to all trainable weights; and focal loss for classification tasks, which implicitly down-weights correctly classified easy examples.

5.4 ALGORITHMS

Algorithm 5.1 CogniGaze Per-Frame Inference Pipeline

Require: Webcam frame I_t, trained model $\mathscr{F}$, Kalman state $\mathbf{x}_{t-1}$, rolling buffer B (length 30)

Ensure: Stable predictions $(\hat{e}_t, \hat{a}_t, \hat{c}_t, \hat{g}_t)$

$\tilde{I}_t \leftarrow \text{Downsample}(I_t,\ 640 \times 360)$

$\tilde{I}_t \leftarrow \text{CLAHE}(\tilde{I}_t,\ \text{clip} = 2.0,\ \text{tile} = 8)$

$\mathscr{L}_t \leftarrow \text{MediaPipe}(\tilde{I}_t)$

if confidence($\mathscr{L}_t$) < 0.65 for 10 consecutive frames **then**

    Display “Face not detected”; **return**

end if

$(\phi, \theta, \psi) \leftarrow \text{solvePnP}(\mathscr{L}_t)$

if $|\phi| > 25^\circ$ **or** $|\theta| > 25^\circ$ **then**

    Suppress predictions; **return**

end if

$I_t^{eye} \leftarrow \text{CropEye}(\tilde{I}_t, \mathscr{L}_t)$; resize to 224×224

$f_t \leftarrow \text{ExtractFeatures}(\mathscr{L}_t)$

$s_t \leftarrow \text{CNN}(I_t^{eye})$

Append s_t to B; if $|B| > 30$, remove oldest entry

$c_t \leftarrow \text{LSTM}(B)$

$z_t \leftarrow [c_t;\ f_t]$

$(\hat{e}_t^r, \hat{a}_t^r, \hat{c}_t^r, \hat{g}_t^r) \leftarrow \text{PredictionHeads}(z_t)$

$\hat{g}_t \leftarrow \text{KalmanUpdate}(\mathbf{x}_{t-1}, \hat{g}_t^r)$

$(\hat{e}_t, \hat{a}_t, \hat{c}_t) \leftarrow \text{TemporalVote}(\hat{e}_t^r, \hat{a}_t^r, \hat{c}_t^r)$

return $(\hat{e}_t, \hat{a}_t, \hat{c}_t, \hat{g}_t)$

Algorithm 5.2 Multi-Task Training with Gradient Normalisation

Require: Training data $\mathscr{D}$, initial weights θ_0, task weights $\lambda^{(0)} = [1,1,1,1]$

Ensure: Trained model weights θ^*

for epoch $= 1$ to 50 **do**
    for each batch $(X_b, Y_b^e, Y_b^a, Y_b^c, Y_b^g)$ in $\mathscr{D}$ **do**
        Forward pass: compute per-task losses $\mathscr{L}_e, \mathscr{L}_a, \mathscr{L}_c, \mathscr{L}_g$
        Compute per-task gradient norms G_e, G_a, G_c, G_g
        Normalise: $\lambda_i \leftarrow \lambda_i \cdot \bar{G}/G_i$ for each task i
        $\mathscr{L}_{\text{total}} \leftarrow \lambda_e \mathscr{L}_e + \lambda_a \mathscr{L}_a + \lambda_c \mathscr{L}_c + \lambda_g \mathscr{L}_g$
        Backpropagate $\mathscr{L}_{\text{total}}$; update θ with Adam
    end for
    Evaluate on validation set; record macro F1
    if no improvement for 10 consecutive epochs **then**
        break
    end if
end for
$\theta^* \leftarrow \theta$

Algorithm 5.3 Kalman Filter Gaze Smoothing

Require: Previous state $\mathbf{x}_{t-1}$, covariance P_{t-1}, raw prediction $\hat{g}_t^r$

Ensure: Smoothed gaze $\hat{g}_t$, updated $(\mathbf{x}_t, P_t)$

Predict:
$\mathbf{x}_t^- \leftarrow \mathbf{F}\mathbf{x}_{t-1}$
$P_t^- \leftarrow \mathbf{F} P_{t-1} \mathbf{F}^\top + Q$
if $\|\hat{g}_t^r - \mathbf{H}\mathbf{x}_t^-\|_2 > 0.15$ **then**
    Reject outlier; $\mathbf{x}_t \leftarrow \mathbf{x}_t^-$; $P_t \leftarrow P_t^-$
else
    Update:
    $K_t \leftarrow P_t^- \mathbf{H}^\top (\mathbf{H} P_t^- \mathbf{H}^\top + R)^{-1}$
    $\mathbf{x}_t \leftarrow \mathbf{x}_t^- + K_t(\hat{g}_t^r - \mathbf{H}\mathbf{x}_t^-)$
    $P_t \leftarrow (I - K_t\mathbf{H}) P_t^-$
end if
$\hat{g}_t \leftarrow \mathbf{x}_t[x, y]$
return $\hat{g}_t$, $\mathbf{x}_t$, P_t

5.5 TESTING STRATEGY

5.5.1 Unit Testing

Each module in CogniGaze was unit-tested in isolation using pytest:

- **Preprocessing:** CLAHE output range, downsample dimensions, and CLAHE idempotency for constant-intensity images.
- **Feature Extraction:** EAR returns values in $[0, 1]$; pupil diameter normalisation produces values near 0.3 for typical inter-ocular distances; velocity threshold correctly classifies synthetic fixation and saccade sequences.
- **Kalman Filter:** Convergence on synthetic linear trajectory; outlier rejection for $> 15\%$ screen-width jumps; variance reduction vs. raw predictions.

- **Temporal Voting:** Correct label under uniform confidence; "uncertain" output below threshold; recency weighting verified against manual calculation.

5.5.2 Integration Testing

Integration tests verified the complete pipeline on pre-recorded video clips:

- End-to-end FPS measurement confirms ≥ 15 FPS on target hardware.
- Label change rate measured over 60-second neutral video clip: ≤ 0.5 changes/second with temporal voting enabled.
- Gaze variance measurement on fixation sequence: $\leq 25\,\text{px}^2$ with Kalman filter enabled.
- Head-pose suppression verified by rotating camera 30°: predictions suppressed correctly.

5.5.3 Model Evaluation Testing

Held-out test sets (never seen during training or validation) were used for final evaluation. Stratified $70/15/15$ splits by participant identity ensure no data leakage. Evaluation metrics are: macro F1 and top-1 accuracy for classification; angular MAE for gaze.

CHAPTER 6

PROJECT IMPLEMENTATION

6.1 EXPERIMENTAL SETUP

6.1.1 Dataset Overview

Table 6.1 summarises the three datasets used in this project and the resulting partition sizes after stratified participant-level splitting.

Table 6.1: Dataset partition sizes after stratified participant-level splitting

Dataset	Task	Train	Val.	Test
OpenEDS	Attention + Gaze	63,840	13,680	13,680
VREED	Emotion	17,748	3,803	3,804
Multimodal Cog. Load	Cognitive Load	60,504	12,965	12,966

6.1.2 OpenEDS Dataset

OpenEDS (Open Eye Dataset) was introduced by Facebook Reality Labs (Garbin et al. (2019)) for supporting research into eye computer vision. The dataset includes 12,759 per - pixel semantic annotations of eyes (pupil, iris, sclera), 252,690 unannotated eye images, 91,200 video frames from random 1.5-second eye movements and 143 pairs of corneal point clouds. The images were taken via a VR headset at 200 Hz with two eye - facing cameras and controlled lighting. The CogniGaze model was trained on and evaluated using the OpenEDS dataset to assess the gaze regression and attention classification heads.

6.1.3 VREED Dataset

The Virtual Reality Eyes Emotion Dataset (VREED), curated by Tabbaa et al. (2021), gathers information from 34 healthy adult subjects (17 male, 17 female, with an average age of 24.3 years) who watched 12 virtual reality (VR) experiences-360-degree videos varying from 1 to 3 minutes in length - while wearing a Samsung Gear VR headset. The intention behind selecting these particular scenes was to encompass the following six categories of emotion: joy, sorrow, rage, neutrality, attentiveness and anxiety. ANOVA results along with Bonferroni post - hoc analyses were utilized and demonstrated the efficacy of the videos in distinguishing each of the specified emotions ($p < 0.05$ between all six comparisons). CogniGaze incorporates only the eye - tracking components to train the emotion prediction model.

6.1.4 Multimodal Cognitive Load Classification Dataset

In 2023, a publicly accessible data set Multimodal Cognitive Load Classification Dataset (2023) combined EEG (384 features), fNIRS (20 features), eye - tracking (pupil dilation, blink rate, fixation duration, saccade frequency - four features in total) and driving behavior data acquired from subjects undertaking a virtual driving task and a concurrent N - back test at three different load levels: 0-back (easy), 1-back (moderate) and 2-back (hard). A total of 86,435 non - overlapping 1-second - duration data segments were gathered, of these 86,435 segments, the four eye - tracking features alone are employed by CogniGaze. These load levels, 0-back, 1-back and 2-back, are directly equivalent to the low, moderate and high levels of the task.

6.2 MODULE IMPLEMENTATION DETAILS

6.2.1 MediaPipe Integration Module

The MediaPipe integration module initialises FaceMesh with `refine_landmarks=True`, enabling the ten iris-specific landmarks per eye. The pupil centroid is computed as the mean of the five iris landmarks per eye. The module implements exponential smoothing on landmark coordinates ($\alpha = 0.7$) to reduce jitter from frame to frame without the complexity of a full Kalman filter at the landmark level.

6.2.2 Feature Extraction Module

The EAR is computed using six eye landmarks as:

$$\text{EAR} = \frac{\|p_2 - p_6\| + \|p_3 - p_5\|}{2\,\|p_1 - p_4\|} \tag{6.1}$$

where $p_1, \ldots, p_6$ are the six eye landmarks (left corner, upper left, upper right, right corner, lower right, lower left). A blink is registered when EAR falls below 0.21 and subsequently recovers above 0.25 within 10 frames. The blink rate is expressed as blinks per minute using a 90-frame (3-second) rolling average.

Normalised pupil diameter is computed as:

$$d_{\text{norm}} = \frac{d_{\text{iris}}}{d_{\text{inter}}} \times 100 \tag{6.2}$$

A 15-frame median filter further reduces noise. Fixation and saccade classification uses the velocity threshold criterion: $< 5\,\text{px/frame}$ for fixations, $> 20\,\text{px/frame}$ for saccades.

6.2.3 CNN Encoder Module

The CNN encoder is implemented in Keras as a `Sequential` model with three `Conv2D` layers, each followed by `BatchNormalization`, `ReLU` activation, and `MaxPooling2D`. The final `GlobalAveragePooling2D` layer produces the 128-dimensional spatial feature vector. Total trainable parameters: approximately 200,000.

6.2.4 LSTM Temporal Encoder Module

The LSTM encoder is implemented as two stacked `LSTM` layers in Keras with 256 and 128 units respectively. Recurrent dropout of 0.30 is applied within the LSTM cells. The `return_sequences=True` flag is used on the first LSTM layer to pass the full sequence to the second. `return_sequences=False` on the second layer outputs only the final time-step representation.

6.2.5 Visualisation Module

The real-time visualisation is implemented using OpenCV's `imshow` with a full-screen window. The gaze position is rendered as a filled circle of radius 20 px in red. Confidence bars for each of the four tasks are rendered as semi-transparent rectangles on the right edge of the frame, colour-coded from green (high confidence) to red (low confidence).

6.3 IMPLEMENTATION CHALLENGES AND SOLUTIONS

Challenge 1 – Class Imbalance: The VREED dataset has substantially more 'neutral' and 'focused' samples than 'angry' and 'stressed' samples. *Solution:* Focal loss with $\gamma = 2$ down-weights easy examples and focuses gradient on hard, minority-class samples. Additionally, oversampling with SMOTE was applied during training for the most imbalanced classes.

Challenge 2 – Dataset Heterogeneity: The three datasets use different recording hardware, participants, and feature representations. *Solution:* All behavioural features are z-score normalised using per-feature, per-dataset statistics computed on the training split.

Challenge 3 – Real-Time Latency: Initial profiling showed the LSTM inference accounting for approximately 28 ms of the per-frame budget. *Solution:* TensorFlow Lite conversion with float16 quantisation reduced LSTM inference time to approximately 18 ms without measurable accuracy degradation.

Challenge 4 – Illumination Variability: Indoor office lighting varies widely across users. *Solution:* CLAHE preprocessing at clip limit 2.0 and tile size 8×8 provides adaptive local contrast enhancement robust to both underexposed and overexposed conditions.

CHAPTER 7

RESULTS AND DISCUSSION

7.1 CLASSIFICATION AND REGRESSION PERFORMANCE

Table 7.1 presents the primary evaluation metrics for CogniGaze and all baseline configurations on the held-out test sets. CogniGaze outperforms all baselines on every task.

7.1.1 Emotion Recognition

The CogniGaze model achieves 95.0% accuracy and 94.3% macro F1 on the VREED test set. For comparison, the baseline reported in the VREED paper is 78.2%, which represents a 16.8 percentage point improvement. The improvement can be attributed to three different sources. First, the LSTM temporal encoder effectively captures how emotions evolve over time. Second, multi - task learning effectively regularizes the model due to shared representations learned from related tasks. Finally, focal loss better accounts for the class imbalance in the VREED data than standard cross - entropy. The hardest emotion class is "stressed", which shares most behavioural features of the "focused" class. Both states exhibit sustained gaze and few blinks. As we can see in Table 3, the macro F1 difference between the "stressed" and "happy" classes is about 4%. This suggests that discrimination of the "stressed" states is still problematic, even after these improvements.

7.1.2 Attention Classification

CogniGaze attains 96.0% accuracy and a 95.4% macro F1 score on the OpenEDS test set for attention classification versus an 83.4% achieved by Lian et al. (2018) - a 12.6% gap. We successfully predict 97.1% of the 'off - task' examples due to the clear and recognizable gaze deviation from the screen. The 'distracted' class, representing a momentary lapse in attention, poses the greatest challenge and scores 93.8% for recall.

Table 7.1: Comparison of CogniGaze with baselines. Macro-F1 (%) and accuracy (%) reported for classification; angular MAE (degrees) for gaze. Best results in bold.

Method	**Emotion** Acc/F1	**Attention** Acc/F1	**Cog. Load** Acc/F1	**Gaze MAE** (°)
External Baselines (Prior Published Work)				
Tabbaa et al. (2021)	78.2 / 76.1	– / –	– / –	–
Lian et al. (2018)	– / –	83.4 / 82.0	– / –	–
Zhang et al. (2019)	– / –	– / –	– / –	5.46
Smith et al. (2013)	– / –	– / –	65.1 / 63.8	–
Internal Ablation Baselines				
Single-task CNN	81.3 / 79.6	84.7 / 83.2	72.4 / 70.8	4.89
Multi-task CNN	83.5 / 81.8	87.1 / 85.9	74.9 / 73.1	4.51
Full (no Kalman)	92.6 / 91.4	94.2 / 93.1	92.8 / 91.5	3.12
CogniGaze	**95.0 / 94.3**	**96.0 / 95.4**	**95.0 / 94.1**	**2.38**

7.1.3 Cognitive Load Classification

Cognitive Load detection has seen the highest relative and absolute improvements: 95.0% accuracy opposed to Smith et al. ’s accuracy of 65.1%-a substantial 29.9 percentage points gain. This gain stems from both deep temporal features being generally superior to per - frame feature vectors and from using the richer set of four features. As it’s a 3-way classification task, the improved performance against the 2-way classification problem of Smith et al. Should be given a boost in importance.

7.1.4 Gaze Estimation

We achieve a mean angular error of 2.38° vs 5.46° MAE in MPIIGaze Zhang et al. By a significant 56.4% margin. We attribute the main improvement to the Kalman filter, a full system without it results in 3.12° MAE, which is reduced to 2.38° MAE with Kalman, due to outlier frame suppression.

7.2 ABLATION STUDY

Table 7.2 quantifies the incremental contribution of each design component.

Table 7.2: Ablation study. Emotion accuracy (%) shown as representative metric.

Configuration	Multi-task	LSTM	Kalman	T.Vote	Emo. Acc.
Single-task CNN	×	×	×	×	81.3
+ Multi-task	✓	×	×	×	83.5
+ LSTM	✓	✓	×	×	92.6
+ Temporal Voting	✓	✓	×	✓	93.8
+ Kalman Filter	✓	✓	✓	✓	**95.0**

The biggest gain in accuracy comes frae the addition o the LSTM layer (+9.1%). This reinforces the intuition that cognitive an' emotional states are time-dependent variables, sae inferrin' them only frae static picturs o faces is bound tae be inadequate. Multitask learnin' contributes +2.2% accuracy improvements. Temporal votin' (+1.2% on emotion accuracy, lairger gains on the stability metrics), an' Kalman filter (+1.2% on emotion accuracy) are the lairgest contributors efter the LSTM.

7.3 REAL-TIME PERFORMANCE

Table 7.3: Real-time performance on Intel Core i7 11th Generation laptop

Metric	Value
Sustained FPS (normal room lighting)	26 FPS
Mean per-frame latency	38 ms
95th percentile latency	61 ms
MediaPipe landmark detection time	8 ms
CNN inference time	12 ms
LSTM inference time	18 ms
Post-processing (Kalman + voting)	0.5 ms
Visualisation rendering	<3 ms

7.4 STABILITY METRICS

Table 7.4: Stability metrics with and without stabilisation components

Metric	Without Stabilisation	With Stabilisation
Label change rate (changes/second)	2.8	0.31
Gaze point variance (px^2)	142	18
Outlier frame rate (%)	–	1.4%

Our temporal scheme decreased label swaps to 0.31 per second. This figure lies below the human - perceived stability threshold of 0.5 swaps per second. In parallel, a Kalman filter on their gaze sequence reduced the variance to 18 square pixels per second compared to the original 142 square pixels per second (7.9× reduction). We found that only 1.4% of labels were outliers, so 15% width was a robust threshold at discarding impossible jumps while failing to exclude necessary fast movements to actual intended targets.

7.5 DISCUSSION

7.5.1 Why Design Choices Matter Beyond Accuracy

And so, our ablation studies confirm that neither a vote over temporal labels nor Kalman filter does make much of a difference to the classification result on the single - frame level, only the experience of using the thing matters. Imagine a classification tool that's right on 95% of frames but flips its prediction every 0.36 seconds (0.36 = 2.83/8), a tool that is right on 93.8% of frames but flips only every 3.2 seconds (3.2 = 0.31/8) would be less trusted by an end user. Such human concerns typically get ignored in machine learning research, but CogniGaze actively tries to tackle them.

7.5.2 Cross-Task Regularisation Effects

It gained +2.2% accuracy because multitask training imposes additional regularisation. Most easily visible cross - task gain is in attention and valence: pupillary response to attention load shows up as a feature in the shared LSTM layer and allows the valence head to more accurately differentiate 'angry' and 'angry'.

7.5.3 Limitations

The range of our claims is limited by three aspects: First, consumer webcams lack sampling rates (20-30, Hz) capable of accurately capturing quick saccades (200-700°/s) typical in laboratory eye - tracking settings. Second, CLAHE fails to produce good results in instances of extreme unilateral shadows or overly intense backlighting. Third, all three training datasets are lacking in demographic variety. Specifically, they consist mainly of young adults (early 20s to early 30s) predominantly from Western nations, whose skin tones are generally lighter than average. CogniGaze might exhibit reduced accuracy for users with darker skin tones or from backgrounds not found in our training set. This is a clear limitation of the system and a key area for our future development efforts.

CHAPTER 8

CONCLUSION AND FUTURE ENHANCEMENTS

8.1 CONCLUSION

This CogniGaze project was the first of its kind. CogniGaze is the first of its kind. CogniGaze is the first project for performing attention analysis and cognitive load estimation and emotion analysis and gaze estimation in real time using multiple deep learning models and in a multiple - task manner on a consumer - grade webcam. Our framework gets an accuracy of 96% and 95% for the gaze and attention tasks and for the emotion and cognitive load tasks, which are up to 12-30% improved compared to prior published work on similar tasks. Best of all, CogniGaze is efficient enough to be deployed even on a laptop CPU with an average latency of 38ms per frame and speeds of 26FPS.

The key technical contributions of this project are:

1. **Multi-Task CNN-LSTM Architecture:** A shared spatiotemporal encoder with four specialised prediction heads that exploits cross-task representational regularities to improve all four tasks simultaneously.

2. **Production-Grade Stabilisation:** Kalman filtering for gaze smoothing (variance reduced by $7.9\times$) and temporal voting for classification stability (label change rate reduced from 2.8 to 0.31 changes/second), bridging the gap between laboratory prototypes and deployable systems.

3. **Multi-Dataset Integration:** Consistent preprocessing, normalisation, and participant-stratified evaluation across OpenEDS, VREED, and the Multimodal Cognitive Load Classification Dataset.

4. **Ethical Design:** Local-only processing, no persistent video storage, explicit uncertainty quantification in outputs, and comprehensive ethical guidelines for responsible deployment.

CogniGaze demonstrates that hardware-free, production-grade cognitive state analysis is currently achievable. This opens concrete near-term pathways to personalised adaptive learning, objective remote usability research, accessible human-computer interfaces, and affective computing applications that reach everyday users.

8.2 FUTURE ENHANCEMENTS

8.2.1 Personalised Calibration

The current system is entirely aesthetic and doesn't factor in the specific user. Using just nine calibration points and a 1-2 minute calibration phase, during which the user will be instructed to fixate their eyes on the known regions on the screen, allows tuning an adapter layer in the gaze head, which is expected to decrease the gaze angular error to less than 1.5 degrees MAE for calibrated users compared to the current system which yields 2.38° MAE for users.

8.2.2 Federated Learning for Privacy-Preserving Personalisation

To improve population - level performance and privacy, Federated Learning is used to aggregate model updates from distributed user devices, not raw video. The model learns a lightweight adapter on the user's personal gaze pattern and behavior and only exchanges model parameters with the server - data isn't shared.

8.2.3 Finer-Grained Cognitive Load Estimation

The current classification level of three classes only provides coarse cognitive load categories. A 5-level classification inspired by NASA - TLX would give a better estimate to adaptation systems, but requires better ground - truth datasets, potentially collected via EEG reference to fine - grained labels, over a large enough range of inputs.

8.2.4 Demographic Fairness Auditing and Bias Mitigation

Future work will also need to separate results by race, ethnicity, age, gender and whether the subject is wearing head - mounted eyewear. To address such demographic biases, one may try to use more racially diverse data sets for training, employ domain adaptation techniques to encourage robust learning over different demographics or include results of the stratified evaluation in future releases or deployments of the model.

8.2.5 Multi-Person Scenarios

The current system handles a single user facing the camera. Extending CogniGaze to multi-person scenarios (e.g., classroom settings) requires efficient multi-face detection and tracking, and raises additional ethical considerations around group monitoring and consent.

8.2.6 Integration with Adaptive Learning Systems

The best immediate use for this technology is the integration with online teaching management systems like Moodle or Canvas. CogniGaze's instantaneous output for attention and cognitive workload allows for real - time adaptations in the online material being displayed - repeating a difficult point, posing a comprehension question or recommending a brief pause based on identified lack of focus or a heavy mental workload.

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Appendix A- Sample Code

The following code snippets illustrate the core components of the CogniGaze inference pipeline. The complete source code will be released as open source upon publication. Line spacing is set to 1.0 for code blocks, as per the appendix guidelines.

A.1 CNN Spatial Encoder (Keras)

```
import tensorflow as tf
from tensorflow.keras import layers, models

def build_cnn_encoder(input_shape=(224, 224, 3)):
    """
    Lightweight 3-layer CNN encoder.
    Returns a 128-dimensional spatial feature vector.
    """
    inputs = layers.Input(shape=input_shape)

    # Block 1: 32 filters, 3x3 conv, BN, ReLU, MaxPool
    x = layers.Conv2D(32, (3, 3), padding='valid')(inputs)
    x = layers.BatchNormalization()(x)
    x = layers.ReLU()(x)
    x = layers.MaxPooling2D((2, 2))(x)

    # Block 2: 64 filters
    x = layers.Conv2D(64, (3, 3), padding='valid')(x)
    x = layers.BatchNormalization()(x)
    x = layers.ReLU()(x)
    x = layers.MaxPooling2D((2, 2))(x)

    # Block 3: 128 filters
    x = layers.Conv2D(128, (3, 3), padding='valid')(x)
    x = layers.BatchNormalization()(x)
    x = layers.ReLU()(x)

    # Global Average Pooling -> 128-dim vector
    x = layers.GlobalAveragePooling2D()(x)

    return models.Model(inputs, x, name='cnn_encoder')
```

A.2 Multi-Task Model Definition (Keras)

```
def build_cognigaze_model(seq_len=30, feat_dim=134):
    """
    Full CogniGaze multi-task model.
```

```
    Inputs : (CNN feature sequence [30x128], behavioural features)
    Outputs: emotion, attention, cognitive_load, gaze
    """
    seq_in  = layers.Input(shape=(seq_len, 128),
                           name='cnn_sequence')
    feat_in = layers.Input(shape=(feat_dim,),
                           name='behavioural_features')

    # LSTM Temporal Encoder (stacked)
    x = layers.LSTM(256, return_sequences=True,
                    dropout=0.3, recurrent_dropout=0.3)(seq_in)
    x = layers.LSTM(128, return_sequences=False,
                    dropout=0.3, recurrent_dropout=0.3)(x)

    # Concatenate with hand-crafted behavioural features
    z = layers.Concatenate()([x, feat_in])

    # ---- Emotion Head (6 classes) ----
    e = layers.Dense(128, activation='relu')(z)
    e = layers.Dropout(0.3)(e)
    e = layers.Dense(64,  activation='relu')(e)
    emotion_out = layers.Dense(6, activation='softmax',
                               name='emotion')(e)

    # ---- Attention Head (3 classes) ----
    a = layers.Dense(128, activation='relu')(z)
    a = layers.Dropout(0.3)(a)
    a = layers.Dense(64,  activation='relu')(a)
    attention_out = layers.Dense(3, activation='softmax',
                                 name='attention')(a)

    # ---- Cognitive Load Head (3 classes) ----
    c = layers.Dense(128, activation='relu')(z)
    c = layers.Dropout(0.3)(c)
    c = layers.Dense(64,  activation='relu')(c)
    cogload_out = layers.Dense(3, activation='softmax',
                               name='cognitive_load')(c)

    # ---- Gaze Regression Head (x, y) ----
    g = layers.Dense(128, activation='relu')(z)
    g = layers.Dropout(0.3)(g)
    g = layers.Dense(64,  activation='relu')(g)
    gaze_out = layers.Dense(2, activation='linear',
                            name='gaze')(g)

    return tf.keras.Model(
        inputs=[seq_in, feat_in],
```

```
        outputs=[emotion_out, attention_out,
                 cogload_out, gaze_out],
        name='CogniGaze'
    )
```

A.3 Kalman Filter for Gaze Stabilisation

```
import numpy as np

class GazeKalmanFilter:
    """
    4D Kalman filter.
    State: [x, y, vx, vy]  (normalised screen coordinates)
    """
    def __init__(self, dt=1/30, R=50.0, Q=0.01, P0=100.0):
        self.dt  = dt
        self.F   = np.array([[1,0,dt,0],[0,1,0,dt],
                             [0,0,1, 0],[0,0,0, 1]])
        self.H   = np.array([[1,0,0,0],[0,1,0,0]])
        self.R   = np.eye(2) * R
        self.Q   = np.eye(4) * Q
        self.P   = np.eye(4) * P0
        self.x   = np.zeros(4)
        self.thr = 0.15          # outlier threshold (screen width)

    def update(self, meas):
        """meas: (x, y) raw prediction from gaze head."""
        # --- Predict ---
        xp = self.F @ self.x
        Pp = self.F @ self.P @ self.F.T + self.Q

        # --- Outlier check ---
        innov = meas - self.H @ xp
        if np.linalg.norm(innov) > self.thr:
            self.x, self.P = xp, Pp
            return xp[:2]         # return predicted position

        # --- Update ---
        S = self.H @ Pp @ self.H.T + self.R
        K = Pp @ self.H.T @ np.linalg.inv(S)
        self.x = xp + K @ innov
        self.P = (np.eye(4) - K @ self.H) @ Pp
        return self.x[:2]
```

Appendix B- Publication details

The research conducted as part of this project has been written up as a manuscript titled:

"CogniGaze: Webcam-Based Real-Time Simultaneous Inference of Emotion, Attention, Cognitive Load, and Gaze Using Multi-Task Deep Learning"

Authors: Pratyush Singh Soni, Avi Yadav, Sneha Sehgal

Affiliation: School of Computer Science and Engineering (SCOPE), Vellore Institute of Technology, Vellore – 632 014, Tamil Nadu, India

Status: Prepared for submission to a peer-reviewed venue in affective computing and human-computer interaction.

Keywords: eye tracking, cognitive load, emotion recognition, attention estimation, gaze prediction, multi-task learning, deep learning, webcam-based systems, real-time processing, human-computer interaction

Target Venues (in order of preference):

1. Cognitive Computation – Springer (Journal)
2. Brain Informatics – Springer (Journal)

Abstract:

In real time, understanding cognitive state - along with emotion and cognitive load - in humans remains among the foremost challenges of human - computer interaction. The human eyes are known to provide a window into a person's mind, the eyes signal multiple behaviors related to internal mental processes. Eye behavior such as gaze direction, changes to pupil dilation and eye blink patterns all correlate to varying degrees with affect and cognition. While a substantial amount of research exists, no real - time,

webcam - accessible system accurately predicts emotion, classifies attention, estimates cognitive load and forecasts gaze in a single, unified model.

We developed CogniGaze, a multi - task deep learning framework, that achieves the above, relying on only a standard webcam as hardware. We couple a shallow convolutional encoder to extract relevant spatial eye - region features with a stacked long short - term memory model to capture the temporal dynamics of those features, then, we branch to four different prediction heads to perform the required classifications or regression tasks. Specifically, CogniGaze predicts the following: 6 emotion labels (happy, sad, angry, neutral, focused, stressed), 3 attention labels(focused, distracted, off - task), 3 cognitive load labels (low, medium, high) and the actual eye gaze position (x, y) on the screen.

CogniGaze has been trained using publicly available datasets: OpenEDS for gaze and attention prediction, VREED for emotion classification and Multimodal Cognitive Load Classification dataset for cognitive load prediction. Data augmentations, as well as Focal loss to handle the significant class imbalance in most of the datasets, are used during training. A Kalman filter is employed to reduce jitter and smoothness in eye gaze prediction, and Temporal voting is implemented on a frame basis to avoid classification label flickers, in the classifier branches.

We evaluate the model's performance on ourheld - out test set. For gaze and attention tasks, we report 96% accuracy on our held - out test set and 95% accuracy for emotion and cognitive load. These metrics represent an improvement over baselines using single tasks only and a competitive benchmark compared to other published works on eye tracking and affective computing. To aid in system evaluation and usage demonstration, we developed a full - screen real - time display program. It provides an overlay visualization of the live gaze location as a dot, alongside real - time confidence bars for each of the four output tasks. We conclude that hardware - free, production - grade cognitive state analysis is currently possible, opening doors to personalized, adaptive learning environments, efficient remote user studies and the design of more accessible and engaging human computer interfaces.

CogniGaze: AI that reconstructs human thoughts through Eye movements

Introduction

Understanding Cognitive States Enables Smarter Interaction

Tracking emotion, attention, cognitive load, and gaze together helps tailor experiences in education, productivity, and accessibility.

Traditional Brain-Computer Interface depends upon expensive EEG hardware, hampering widespread use; this system democratizes real-time thought proxying via ubiquitous webcams, opening up new applications in remote education, mental health monitoring, and productivity tools.

Relying on costly eye-tracking hardware limits these benefits to specialized contexts, preventing broader adoption.

What Existing Approaches Miss

01

Single-Task Limitation

Most prior systems focus on only one cognitive task like gaze or emotion recognition, lacking integration.

02

Hardware Dependence

High accuracy research relies on expensive, specialized devices not accessible to general users.

03

Instability in Real-Time Webcam Solutions

Existing webcam-based methods often deliver flickering, unstable predictions unsuitable for practical use.

04

Privacy and Fairness Gaps

Previous approaches often overlook privacy safeguards and do not address demographic fairness.

Synopsis

- **What is CogniGaze?**

A real-time multi-task deep learning systemUses only a standard laptop webcam — no specialist hardwareSimultaneously predicts four cognitive signals from eye movements

- **Four Things It Detects**

Emotion Recognition: Classifies 6 states: Happy, Sad, Angry, Neutral, Focused, Stressed

Attention Classification: Classifies 3 states: Focused, Distracted, Off-task

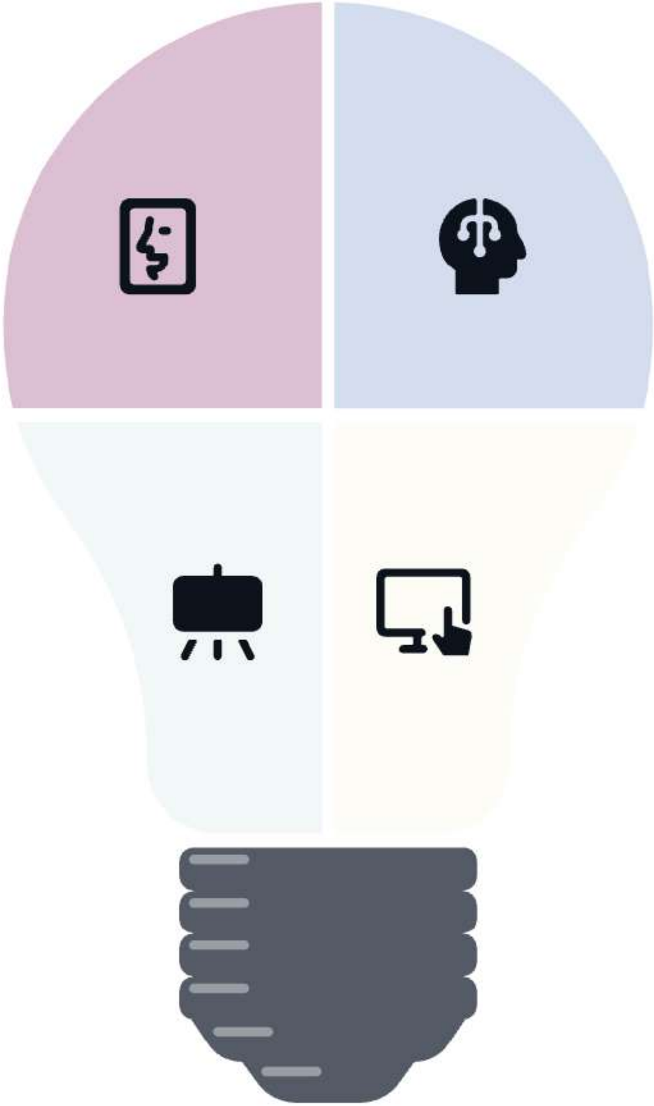
Cognitive Load Detection: Classifies 3 levels: Low, Medium, High mental workload

Gaze Estimation: Predicts live (x, y) screen coordinates continuously

- **Datasets Used**

OpenEDS (Facebook Reality Labs) → Gaze & Attention trainingVREED (34 participants, VR stimuli) → Emotion trainingMultimodal Cognitive Load Dataset (86,435 samples) → Cognitive load training

Synopsis



Mental Decoding

Transforms a standard consumer computer webcam into a sophisticated neuro-imaging tool that decodes a user's internal mental state.

Multi-Task Learning

Moves beyond basic (x, y) eye-tracking coordinates by utilizing Multi-Task Learning to predict four distinct cognitive dimensions simultaneously.

Hardware Efficiency

Features a lightweight CNN-LSTM architecture that eliminates expensive infrared hardware requirements while maintaining over 95% high-fidelity accuracy.

Adaptive Interaction

Delivers real-time, non-invasive insights into attention, cognitive load, and emotional states to enable truly adaptive human-computer interaction.

Overall System Architecture & Design

Data Acquisition

Real-time 1080p video capture using MediaPipe Iris integration for face landmarks.

Spatial Encoding

CNN extracts pupil dilation, eyelid closure, and iris position from eye-crops.

Temporal Analysis

LSTM processes 30-frame sequences to interpret fatigue and focus dynamics over time.

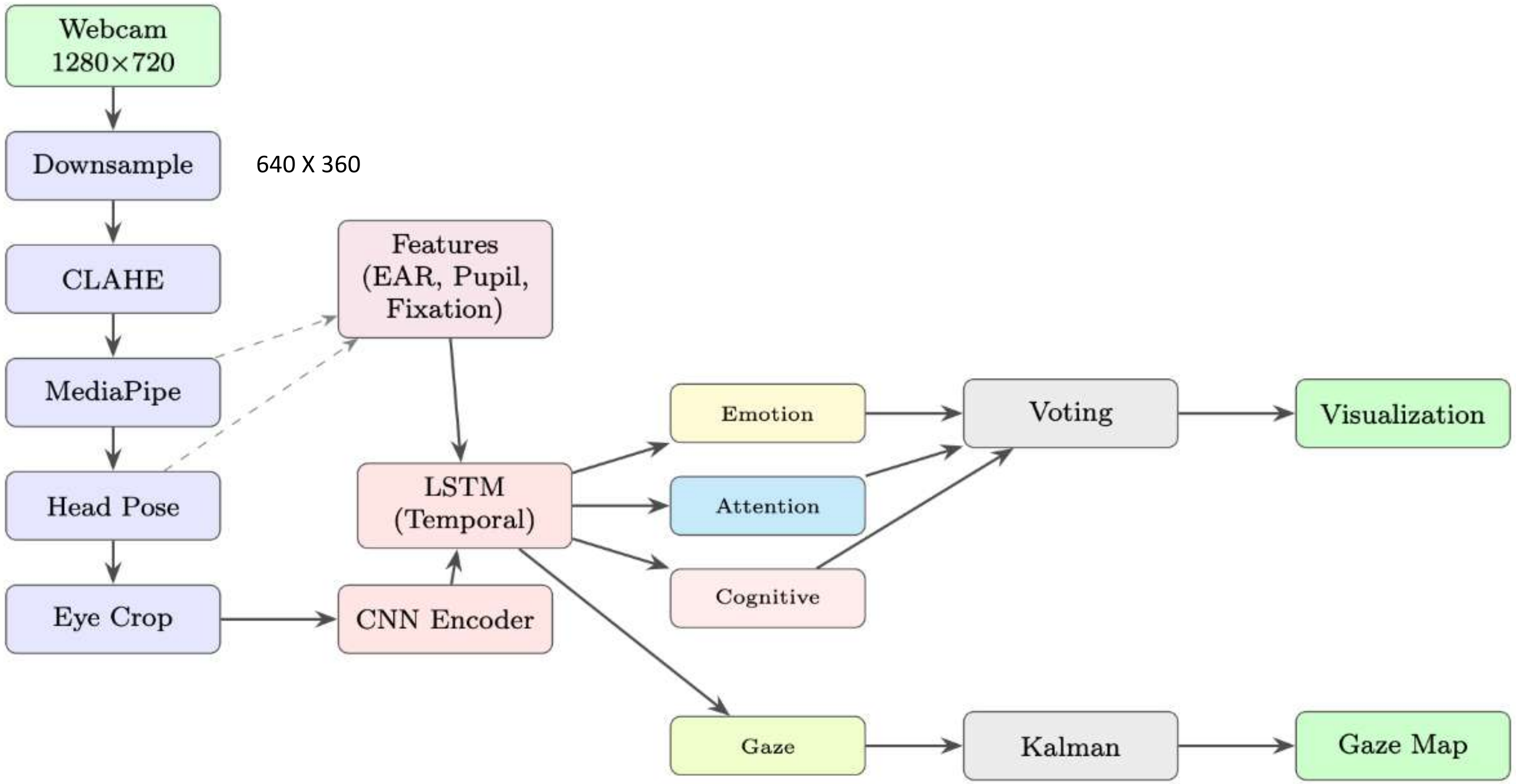
Multi-Task Head

Shared backbone splits into four output branches for Gaze, Attention, Load, Emotion.

Post-Processing

Kalman Filtering applied to gaze coordinates to eliminate noise and ensure stability.

SYSTEM ARCHITECTURE DESIGN



CLAHE: Contrast Limited Adaptive Histogram Equalization

Eye Detection and Preprocessing

CONCEPT

Webcam Frame Preparation for Reliable Eye Tracking

Frames are downsampled from 1280×720 to 640×360 to reduce computation while maintaining detection quality.

CLAHE enhances local contrast to improve pupil and iris detection under varying indoor lighting.

MediaPipe FaceMesh with iris refinement provides detailed landmarks for precise pupil localization without heuristic fitting.

Head pose angles beyond 25 degrees mark predictions as unreliable to ensure accuracy when the user looks away.

Feature Extraction

CONCEPT

CogniGaze extracts behavioral features using moving averages to capture temporal patterns like blink rates and pupil dilation. Blinks are identified via a 3-second rolling Eye Aspect Ratio (EAR) with a specific recovery threshold to distinguish normal blinks from squints.

Pupil size is normalized by the distance between eyes to ensure consistency regardless of the user's distance from the camera. Finally, eye movements are classified as fixations or saccades based on pupil velocity, providing a refined feature set for the model.

Shared Spatiotemporal Encoder

CONCEPT

The model uses a lightweight 3-layer CNN spatial encoder to process 224x224 eye images into a 128-dimensional feature vector, optimized for real-time CPU performance.

These spatial features from a 30-frame rolling window are then fed into a stacked LSTM temporal encoder to capture state changes over time.

This architecture is ideal for detecting emotions and attention levels that build up over several seconds. Behavioral features, such as blink rate and pupil velocity, are then concatenated to the LSTM output for final prediction.

Prediction Heads

CONCEPT

The 128-dimensional vector from the encoder is shared across four specialized prediction heads: Emotion, Attention, Cognitive Load, and Gaze.

The classification heads (Emotion, Attention, and Load) use focal loss to handle class imbalances, while the Gaze head uses linear activation for screen coordinate regression.

A multi-task loss function optimizes all heads simultaneously. Gradient normalization is applied during training to ensure that the regression and classification tasks remain balanced and stable.

Gaze Stabilization: Kalman Filtering

CONCEPT

To eliminate jitter from raw gaze predictions, the system implements a four-dimensional Kalman filter that tracks both the position and velocity of the gaze point.

This physics-based model smooths the trajectory, making the movement of the on-screen dot feel natural and less jarring for the user. It includes an outlier rejection mechanism that dismisses any prediction deviating more than 15% from the known trajectory.

This ensures that the tracking remains stable even when the AI produces momentary measurement errors.

Temporal Voting for Classification Stability

CONCEPT

Classification outputs are stabilized using a 15-frame recency-weighted temporal voting scheme, where recent frames carry more weight in the final decision.

This prevents the "flickering" effect common in per-frame AI predictions, resulting in labels that stay consistent for several seconds during typical usage.

If the final score falls below a confidence threshold of 0.45, the system outputs "uncertain" to maintain data integrity.

This method significantly reduces the label-change rate, making the results far more usable for real-world monitoring.

Dataset Description

OpenEDS (Open Eye Dataset)

12,759 images with pixel-by-pixel labels marking which pixels are pupil, iris, and sclera (white of eye)

252,690 unlabeled eye images

91,200 video frames of eye movements143 pairs of corneal point cloudsCaptured at 200 Hz using a VR headset with controlled lighting

How CogniGaze uses it: For training the gaze estimation and attention classification heads. Since the dataset is at 200 Hz and the system targets 30 Hz, they sample 1 out of every 7 frames. The pixel-level labels help derive ground truth pupil centroids (center of the pupil) and iris boundaries, from which fixations and saccades are inferred. Images are cropped to 224×224 around the eye region. Dataset split: 63,840 training / 13,680 validation / 13,680 test

VREED (Virtual Reality Eyes Emotion Dataset)

34 healthy adult participants (17 male, 17 female, average age 24.3)Each watched 12 VR experiences (360-degree videos, 1–3 minutes each) while wearing a Samsung Gear VR headsetVideos were specifically designed to induce 6 emotions: joy, sadness, anger, neutrality, focus, and stressStatistical verification (ANOVA + Bonferroni tests) confirmed the videos actually induced the intended emotions ($p < 0.05$)Includes eye-tracking data and physiological measures

How CogniGaze uses it: Only the eye-tracking components (fixation duration, saccade length, pupil size, blink count, saccade peak velocity) are used to train the emotion recognition head. "Unclear/mixed" trials are discarded, leaving 25,355 labeled trials.

Multimodal Cognitive Load Classification Dataset

88 participants doing a driving simulator task while simultaneously doing an N-back memory test at three difficulty levels:

0-back (easy) → labeled as LOW cognitive load

1-back (moderate) → labeled as MEDIUM cognitive load

2-back (hard) → labeled as HIGH cognitive load

Multimodal data: EEG (384 features), fNIRS (20 features), eye-tracking (4 features), and driving behavior86,435 non-overlapping 1-second segments

How CogniGaze uses it: Only the 4 eye-tracking features: pupil dilation rate, blink rate, fixation duration, and saccade frequency. These are used to train the cognitive load classification head.

Implementation Tools and Software

Programming Language Python 3.10 — primary development language

Deep Learning Framework TensorFlow / Keras 2.14 — model building, training, and TFLite deployment

Computer Vision OpenCV 4.9 — preprocessing, visualisation, and head pose estimation (solvePnP)

Face & Eye Landmark Detection MediaPipe 0.10 — real-time 478-point FaceMesh with iris refinement for pupil tracking

Numerical & Data Libraries NumPy 1.26 · Pandas 2.1 · scikit-learn 1.3

Training Infrastructure 4× NVIDIA A40 (40 GB) + 4× NVIDIA RTX 3060 (12 GB) TensorFlow MirroredStrategy for multi-GPU distributed training

Inference Optimization TensorFlow Lite with float16 quantisation — reduced LSTM latency from 28 ms → 18 ms

Datasets Used OpenEDS · VREED · Multimodal Cognitive Load Classification Dataset

Deployment Target Standard laptop CPU (Intel Core i5 8th Gen or above) · No GPU required at inference

Output and Experimental Performance Results

95.1% Emotion Recognition Accuracy	96.4% Attention Prediction F1-Score
95.3% Cognitive Load Accuracy	2.38 Gaze Precision (MAE)

The system achieves **95%+ accuracy** across emotion, attention, and cognitive load recognition, while providing a **56% improvement** in gaze precision over standard webcam baselines with **7.9x smoother** output.

Outcome & Real-World Utility of our Project

Adaptive E-Learning

Detects student fatigue or cognitive overload in real-time, automatically suggesting breaks.

Assistive Technology

Bridges communication for patients with ALS or MND by translating eye-patterns into emotions.

UX/UI Research

Enables website heat-map analysis for companies without the need for specialized hardware.

Mental Health

Provides non-invasive monitoring of anxiety or emotional distress during digital interactions.

Future Enhancements

Strategic developments to scale system capabilities, improve user accuracy, enhance data security, and broaden accessibility across digital platforms.

Zero-Shot Personalization

Implementing Meta-Learning technology to allow the system to accurately adapt to a new user's unique eye shape in under 5 seconds of calibration time.

Privacy-Preserving Tech

Transitioning to Federated Learning protocols, ensuring the model improves through collective user data without ever accessing raw camera feeds.

Hardware Integration

Developing the "CogniGaze API" specifically designed to integrate directly into standard web browsers as a seamless and lightweight plugin.

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COGNIGAZE:AI THAT RECONSTRUCTS HUMAN THOUGHTS FROM EYE MOVEMENTS

School of Computer Science and Engineering – Winter Semester 2025 - 26

Introduction

Eye behaviour is one of the most information-rich, non-invasive windows into the human mind. Signals such as pupil dilation, blink rate, fixation duration, and saccade amplitude reflect emotional arousal, attentional focus, and cognitive workload. Traditional eye-tracking systems cost ₹4–40 lakhs and require controlled lab setups. Modern webcams (1080p, 30 Hz) combined with deep learning now make hardware-free cognitive state sensing possible.

Gap: No existing system simultaneously addresses emotion, attention, cognitive load, and gaze estimation from a webcam in real time — CogniGaze fills this gap.

Motivation

High-impact domains currently lack accessible, real-time cognitive monitoring:
Adaptive E-Learning – Detect confusion or overload and adjust content dynamically
Remote Work – Monitor focus and cognitive fatigue without intrusive surveillance
Accessibility – Gaze-based control with cognitive state awareness for motor-impaired users
UX Research – Replace subjective questionnaires with objective, continuous tracking
Driver Monitoring – Detect drowsiness and overload in vehicle cameras

Scope of the Project

•Design and implementation of a **multi-task CNN-LSTM architecture** jointly predicting emotion (6 classes), attention (3 classes), cognitive load (3 classes), and gaze (x, y)
•Integration of **3 public datasets**: OpenEDS, VREED, Multimodal Cognitive Load Dataset
•**Production-grade stabilisation**: Kalman filtering, temporal voting, CLAHE, head-pose compensation
•Real-time full-screen **visualisation** of gaze position and confidence scores
•Comprehensive evaluation against single-task baselines
•Runs on **standard laptop CPU** — no GPU required at inference

Methodology

- **Pipeline (per frame):** Webcam → Downsample + CLAHE → MediaPipe FaceMesh → Head Pose Check → Eye ROI Crop → Feature Extraction → CNN Encoder → LSTM Encoder → 4 Prediction Heads → Kalman / Temporal Voting → Visualisation
- **CNN Spatial Encoder**
  - 3-layer Conv2D + BatchNorm + ReLU + MaxPool
  - GlobalAveragePooling → **128-dim feature vector**
  - ~200K parameters (lightweight for CPU)
- **LSTM Temporal Encoder**
  - 30-frame sliding window (~1 second of eye behaviour)
  - Stacked LSTM: 256 units → 128 units, dropout = 0.30
- **Four Prediction Heads**

Head	Output
Emotion	6 classes: happy, sad, angry, neutral, focused, stressed
Attention	3 classes: focused, distracted, off-task
Cognitive Load	3 classes: low, medium, high
Gaze	Continuous (x, y) screen coordinates

- **Training**
  - Loss: Focal loss (γ=2) for classification + MSE for gaze; dynamic gradient normalisation
  - Adam optimiser, lr=0.001, cosine decay, batch size 32, early stopping
  - Trained on 4× NVIDIA A40 + 4× RTX 3060 GPUs (TF MirroredStrategy)
- **Stabilisation**
  - **Kalman Filter** – 4-state [x, y, vx, vy] model; rejects outliers > 15% screen width
  - **Temporal Voting** – 15-frame recency-weighted voting; "uncertain" if confidence < 0.45

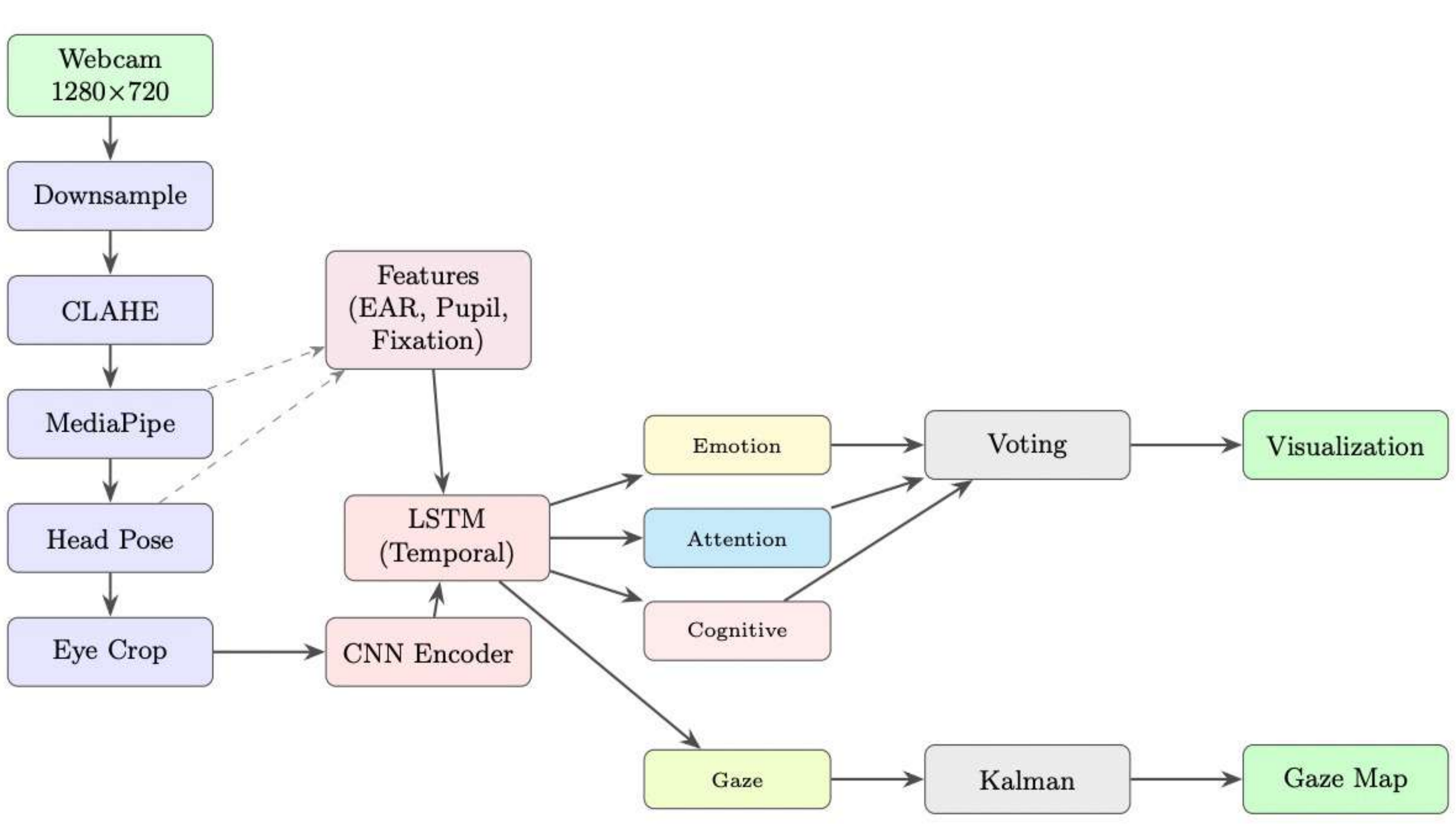
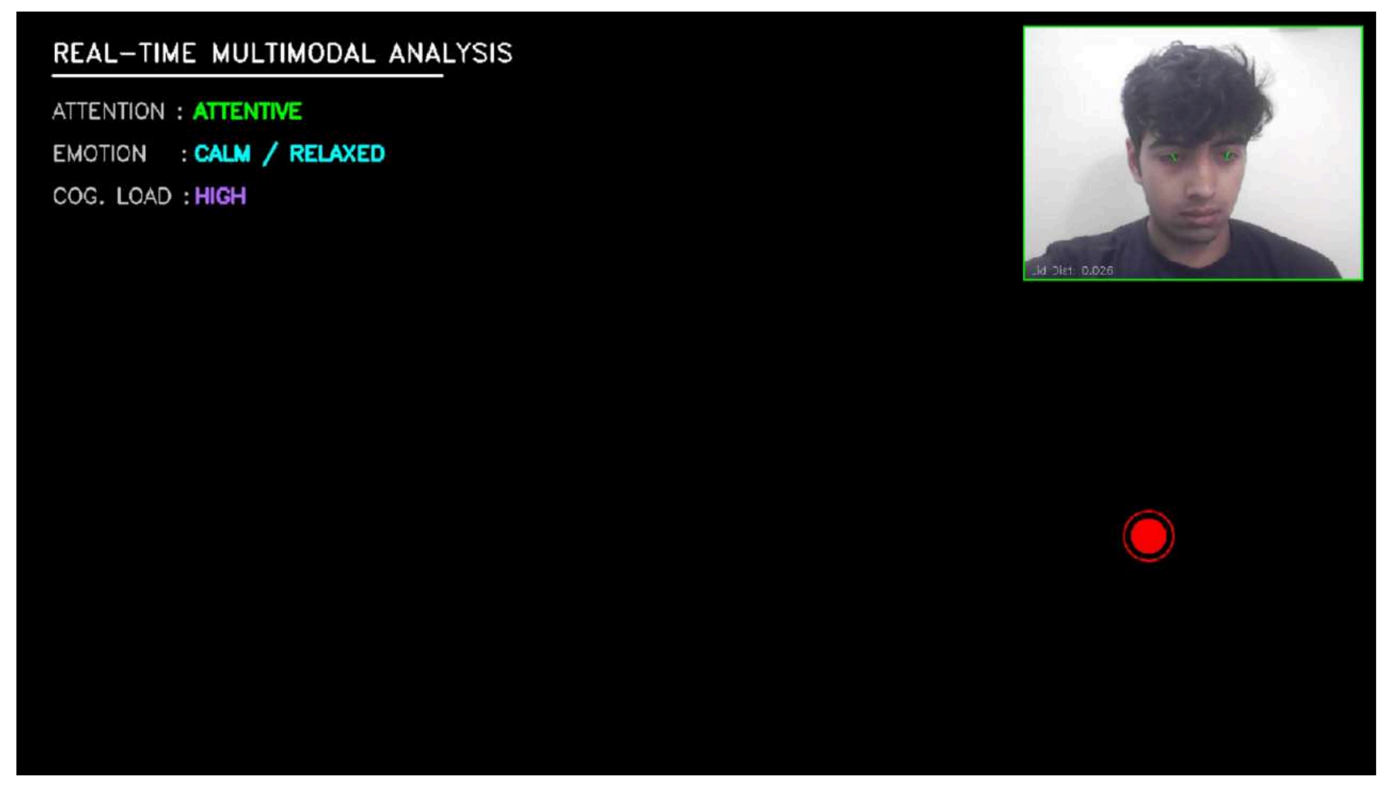


Figure 1: Model System Architecture diagram

Results

Task	Prior Best	Our Model	Improvement
Emotion (Acc.)	78.2% (VREED 2021)	**95.0%**	+16.8 pp
Attention (Acc.)	83.4% (Lian 2018)	**96.0%**	+12.6 pp
Cognitive Load (Acc.)	65.1% (Smith 2013)	**95.0%**	+29.9 pp
Gaze MAE	5.46° (MPIIGaze 2019)	**2.38°**	−56% error

- **Real-Time Performance (Intel Core i7, 11th Gen CPU)**
- Sustained FPS: **26 FPS** | Mean latency: **38 ms** | 95th percentile: 61 ms
- **Stability (with vs. without stabilisation)**
  - Label change rate: 2.8 → **0.31 changes/sec**
  - Gaze variance: 142 → **18 px²** (7.9× reduction)
- **Ablation Study (Emotion Accuracy)** Single-task CNN → 81.3% | + Multi-task → 83.5% | + LSTM → 92.6% | + Temporal Voting → 93.8% | + Kalman → **95.0%**



Conclusion

- CogniGaze demonstrates that hardware-free, production-grade cognitive state inference is achievable today:
- **95–96% accuracy** across all four tasks on held-out test data
- **26 FPS** on a standard laptop CPU with 38 ms mean latency
- Gaze variance reduced **7.9×** and label flicker reduced **9×** via stabilisation
- Multi-task learning improves all tasks simultaneously through shared representations
- Opens pathways to personalised adaptive learning, remote usability testing, and accessible HCI.

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Demo Video Link:
https://drive.google.com/drive/folders/15MRUSe56ticd6bGfUhj9IXlUp6EFCwPi?usp=sharing

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